An Evidence-Gated, Fail-Closed Pre-Board Contract for Drift-Adaptive GKP Decoding

Evidence-synchronized manuscript note; snapshot through T7.2.4, 20 July 2026

Abstract

Approximate Gottesman–Kitaev–Preskill (GKP) quantum error correction must decode analog modular syndromes while the effective displacement, measurement, and control statistics drift. We reorganize an earlier CNN-assisted, FPGA-oriented dual loop as the Route-A contract system: a locked MAP fast path owns logical-error decisions, observed-only software logic owns bounded adaptation, and an event/state-machine layer governs fail-closed publication to versioned A/B banks. CNN, teacher, student, and HMM inference remain optional software extensions rather than the deployable claim.

Route-A passed preregistered simulator gates and a one-million-cycle CXXRTL pre-board replay with zero visible-field mismatches. On the preregistered smooth matrix, it lowered aggregate paired logical error rate relative to the locked EWMA baseline by 2.1687×10^{-5} (95% interval $[1.9003, 2.4548] \times 10^{-5}$); periodic drift was the only family retained after Holm correction. The selected no-student hardware profile preserved a six-cycle, initiation-interval-one architecture and passed three-seed open-source placement and routing at the 27-MHz contract clock. Frequency, resource, and power values are post-route estimates rather than board measurements. These results support a restricted simulator/pre-board contract paper whose contribution is deterministic, evidence-gated integration, not an unqualified decoder or hardware ranking.

Contents

1 Introduction

The Gottesman–Kitaev–Preskill (GKP) code protects oscillator-encoded quantum information by converting small phase-space displacements into analog modular syndromes [1, 2]. Practical code states have finite energy, so peak width, envelope, loss, and the recovery map affect the induced logical channel rather than merely perturbing an ideal lattice decision [3, 4]. Autonomous stabilization and real-time GKP error correction have further made feedback timing and failure handling physical constraints, not optional implementation details [5, 6]. The classical path in a repeated GKP memory must therefore be assessed by what it observes, what action it controls, and when that action becomes valid.

Neither analog decoding nor temporal inference is new. Analog likelihoods have improved concatenated surface–GKP and QLDPC–GKP decoding, maximum-likelihood (ML) rules outperform closest-integer (CI) decisions in matched gate models, and multi-round Bayesian inference can exploit approximate-GKP syndrome history [7, 8, 9, 10, 11]. Closest-point decoding (CPD) also gives a principled formulation for structured multimode GKP lattices [12]. These results establish the value of soft, correlated, and historical information. They do not, however, make CI, gate-level ML, single-mode MAP, and multimode CPD interchangeable: their code spaces, observations, actions, and performance denominators differ.

Nonstationarity introduces a second problem. Syndrome statistics have been used to estimate time-varying error weights, infer noise parameters, calibrate experimental decoders, reweight decoding graphs, and optimize decoder priors [13, 14, 15, 16, 17]. Learned decoders and calibration-conditioned models provide complementary ways to absorb complex or changing device statistics [18, 19]. In parallel, FPGA QEC decoders have demonstrated low-latency lookup, distributed, clustering, and neural implementations, including processor-integrated feedback [20, 21, 22, 23, 24]. Yet an algorithmic latency, an FPGA core latency, and a source-to-action closed-loop latency are different observables; a numerical comparison is meaningful only after the task and boundary are matched.

The source set audited in this work leaves a narrower systems question. For a repeated single-mode approximate-GKP memory under smooth drift and abrupt fault regimes, can observed syndrome and health information update a decoder without giving a slow estimator authority over the cycle-critical action? Answering this question requires an execution contract as well as an estimator: the contract must define causal observability, expert eligibility, atomic state publication, fallback and rollback semantics, latency boundaries, and the evidence needed to promote a simulator result to hardware. It must also retain negative results when adaptation has insufficient causal action-space headroom.

Here we study that question through the Route-A contract system. A locked MAP expert owns the logical-error decision; observed-only Window, EWMA, Kalman, or regime logic may propose bounded updates; typed event and fallback logic owns tail safety; and a versioned A/B bank with last-known-good rollback owns state publication. The six-cycle, initiation-interval-one fast path executes only validated integer state. CNN residuals, hidden-state models, and recurrent teacher–student modules remain replaceable software extensions rather than the authority or identity of the deployed decoder.

The present evidence supports a restricted conclusion. On the frozen smooth matrix, Route-A improves aggregate paired logical error rate only relative to the locked EWMA baseline, with periodic drift the sole Holm-confirmed family; static joint MAP has a lower average error rate, and Window MAP remains a stronger counterexample. Abrupt and out-of-distribution tests establish the registered non-inferiority margins rather than broad tail improvement. A planned posterior-predictive V5 branch was stopped before formal or RTL work because its causal deployable headroom gates failed. The retained hardware result is consequently a deterministic pre-board contract and post-route estimate, while physical-board latency, jitter, power, and any claim of being faster than prior FPGA decoders remain unmeasured. The contribution is thus an evidence-gated integration and falsification framework: it separates LER, safety, and deterministic execution owners, preserves task-local positive results, and prevents heterogeneous evidence from becoming a global ranking.

2 Scope, argument, and evidence cutoff

The one-sentence argument of this note is:

> For repeated approximate-GKP correction, the ROUTE-A contract system integrates a locked MAP fast path with observed-only adaptation and fail-closed state publication, supported by restricted simulator and pre-board evidence; every performance, auxiliary, and hardware statement remains bounded by its registered task and evidence layer.

The old task board and `docs/paper_notes/CNN_FPGA_GKP_theory_note_draft.tex` remain the historical source for the CNN+FPGA affine-calibration line. The current claim state is instead governed by `docs/new_task_board.md`. This note preserves useful old results as extension evidence without allowing them to override newer matched-comparison failures. The physical setting follows analog GKP syndrome decoding [1, 2]; the adaptive-decoder motivation is aligned with evidence that decoder calibration and prior mismatch affect logical performance [15, 17].

Table 1: Evidence cutoff used in this note. "Done" means that the registered task produced a terminal artifact; it does not imply a positive verdict.

Surface	Status at cutoff	Consequence for this note
T6.5.1–T6.9.3 (V4)	Done except T6.9.2 Blocked	V4 passed restricted simulator/pre-board gates but ended `NO_GO_FULL_HIGH_LEVEL_PAPER`; real-board fields remain null.
T6.10.1 / T6.15.5 (V5)	Done	Causal-headroom entry gate failed and V5 ended `NO_GO_V5_EARLY_HEADROOM_STOP`; no V5 formal, RTL, or P&R evidence exists.
T6.16.1–T6.19.3 (Phase 6C)	Done	The 206-cell six-lane atlas passed 24/24 integrity gates; literature/reproduction/project-native lanes remain separate and cannot upgrade V4/V5.
T7.1.1	Done	The 29-claim matrix passed 19/19 gates and mutations with 45 live artifacts; the manuscript decision is `RESTRICTED_PREBOARD_CONTRACT_PAPER`.
T7.1.2	Done	Main Figures 1–2 bind 38 elements to parent claims and evidence, passed 16/16 gates and mutations, and preserve V5-dropped and board-null surfaces on the figure face.
T7.1.3	Done	Main Figures 3–4 bind 55 result/hardware rows to live parent evidence, passed 16/16 gates and mutations plus 153 parent-chain regressions, and passed manual visual QA.
T7.1.4	Done	Supplementary Figures S1–S5 bind 792 heterogeneous records to 13 live parent verifiers, passed 17/17 gates and mutations plus 210 environment-aware parent regressions, and preserve failure, incomparable, dropped, and board-null states.
T7.2.1–T7.2.3	Done	Introduction/Related Work, Methods, and Results are governed by 14 source rows, 18 method-state rows, and 27 result-state rows; the three prose contracts pass 18/18, 18/18, and 20/20 gates with matched semantic mutations.

2.1 Terminology ledger

The following names are used consistently throughout the note.

2.2 Paper-level claim placement contract

T7.1.1 converts the accumulated evidence into 29 atomic manuscript claims. Each row binds safe English wording, a Chinese boundary, allowed manuscript locations, upgrade or revocation conditions, and report/raw/code hashes. The matrix is therefore a publication contract rather than a post-hoc summary.

The terminal manuscript decision is `RESTRICTED_PREBOARD_CONTRACT_PAPER`; the broader cross-lane paper remains `NO_GO`. This decision also explains why independent Phase 6C advantages are retained below but removed from the abstract. T7.1.2 consumes the same rows in 38 figure elements rather than introducing new claims.

3 Related Work

3.1 Analog, history-aware, and structured GKP decoding

Hard CI discards distance-to-boundary information that is retained by a wrapped likelihood. Analog GKP information has consequently been incorporated into surface-code matching and QLDPC message passing, while gate-level ML can use correlated shift histories that a componentwise CI rule ignores [7, 8, 9, 25, 11]. For an approximate single-mode GKP memory, a Bayesian memory-assisted decoder instead uses the sequence of modular outcomes to infer the current correction [10]. Structured multimode lattices lead to another decision object: CPD identifies a lattice point or coset, with complexity determined by lattice structure [12]. We therefore use separate task-signature lanes for single-mode MAP, two-GKP CNOT CI/ML, and multimode CPD. The project reproduction that CI and Euclidean CPD coincide for the isotropic square single-mode case is a special-case equivalence, not a class-wide decoder ranking.

3.2 Calibration and drift-adaptive decoding

Adaptive QEC includes several distinct operations. Sliding or weighted windows can update matching weights under time-dependent noise [13]; syndrome-only estimators can identify stationary noise parameters [14]; experimental calibration can reconstruct correlated decoding graphs [15]; and graph reweighting or direct prior optimization can target decoder mismatch rather than physical-parameter error [16, 17]. Calibration-conditioned neural models likewise separate slowly varying device context from the fast syndrome path [19]. These studies motivate our observed-only update constraint and the inclusion of Window, EWMA, Kalman, and change-point baselines. Our distinctive unit is not a new estimator family, but the contract that limits when an estimate may alter an action bank and records false update, fallback, recovery, and tail costs separately from average LER.

3.3 Learned, non-Markovian, and autonomous feedback

Direct neural decoding maps a syndrome history to a correction, as demonstrated for surface and bosonic-code settings [18, 26]. Model-free reinforcement learning instead changes a control policy, whereas the non-Markovian feedback (NMF) method of Puviani *et al.* uses model-based Feedback-GRAPE and a recurrent controller to choose future physical-control parameters from the measurement history [27, 28]. An autonomous GKP protocol is different again: engineered dissipation and reset implement physical QEC without a per-round classical syndrome decoder [5]. We keep these decision objects separate. The NMF paper motivates a teacher extension, but the available official repository did not support a paper-exact matched reproduction; existing project checkpoints yield zero learned/controller entries eligible for the frozen same-task ranking. Neither fact is evidence that the contract system surpasses NMF, RL, direct neural decoding, or autonomous QEC.

Table 2: Canonical terminology and authority boundaries.

Canonical term	Definition	What it does not mean
ROUTE-A contract system	Locked MAP fast path, observed-only routing, event/FSM safety, versioned A/B banks, and deterministic execution	A synonym for the CNN, teacher, student, HMM, or oracle
Static joint MAP	Observed-only strong software comparator using a fixed noise model	A universal GKP MAP decoder; the current LUT is source-bound
Window MAP / EWMA adaptive MAP	Observed-only adaptive MAP experts updated from completed causal windows	Hidden-state or per-scenario oracle tuning
Observed-only posterior-predictive weighted CPD	Independent multimode secondary-lane decoder	A Route-A/V5 result or a cross-task comparator
Regime posterior	Causal software evidence used to route between validated expert banks	An FPGA-resident HMM, direct correction, or privileged truth channel
Tail-safety layer	Event FSM, health flags, fallback, leakage/reset handling, and integrity rollback	Tail-performance superiority; T6.7.2 establishes only locked-EWMA non-inferiority
Source-to-action latency	Time from accepted syndrome input to committed action at a stated boundary	A synonym for decoder-core or end-to-end QPU latency
Fast path	Quantized phase-LUT, event, frame, and bank action with a six-cycle contract	A board measurement
Legacy CNN residual	Historical bounded learning ablation behind the same contract	The required main-system winner
Hidden-state oracle	Nondeployable model reference evaluated in an isolated truth lane	A deployable comparator or a paper-level recovery optimum
V5 candidate family	A proposed continuation admitted only if causal deployable headroom clears the preregistered entry gate	An implemented system; V5 was stopped before formal/RTL work
Secondary evidence lane	A task-local reproduction or project-native matched result with an explicit evidence grade	Evidence that can rescue or reopen the V4/V5 promotion gate
Post-route estimate	P&R-derived timing, resources, or power without physical-board execution	Board measurement, transport, or HIL
Board measurement	Result produced by the named physical bitstream, transport, clock, and measurement protocol	Any simulator, CXXRTL, synthesis, or post-route estimate

Table 3: Terminal claim-placement state at the T7.1.1 cutoff. Negative, blocked, and prohibited rows remain part of the result rather than being silently omitted.

Publication state	Count	Manuscript consequence
Allowed restricted	4	Only three claims enter the abstract: contract-system integration, smooth locked-EWMA contrast, and deterministic pre-board architecture.
Results only	7	Task-local quantitative results remain in Results or Supplement and do not enter the title.
Mandatory negative	2	Static-MAP superiority and zero eligible learned checkpoint remain visible in Results/Conclusion.
Blocked / prohibited positive	13	V5, board-measured, GQF/NMF, and speed-superiority claims remain absent as positive results and explicit as limitations.
Meta / related-work only	3	Auxiliary-lane and FPGA-literature rows constrain provenance and placement but do not create a ranking.
Total	29	45 live artifacts; 19/19 integrity gates and 19/19 substantive mutations passed.

3.4 Deterministic and FPGA QEC decoders

FPGA QEC work spans compressed lookup tables, distributed union–find, collision/local clustering, belief propagation, and quantized neural inference [20, 21, 22, 29, 30, 24]. The literature demonstrates that useful hardware evidence must name the code, problem size, device, precision, throughput or initiation interval, latency boundary, and measurement layer. Processor-integrated experiments further separate decoder-core time from communication and feedback response [23, 24]. Our source normalization found no external FPGA implementation with the same single-mode analog input, MAP/event action, precision, and boundary. A count of zero same-task comparators prevents a fair speed ranking; it does not imply that the project is faster. We accordingly report six cycles and initiation interval one as an RTL contract, and frequency and resources as three-seed post-route estimates until the named board is measured.

3.5 Position of the present work

The component ideas in ROUTE-A—analog MAP, causal estimation, learned extensions, event handling, double-buffered state, and deterministic FPGA execution—all have relevant precedents. The earlier CNN-centric prototype also established a useful slow-estimator/fast-rule decomposition, but its learned branch did not survive the later unified evidence gate: no matched learned checkpoint was eligible, no family-level learned discovery survived the registered correction, and one calibration transient worsened from 37 to 55 errors per 512 decisions. The present paper therefore makes a systems and evidence claim. MAP owns LER; typed events, fallback, and rollback own safety; the fast path owns deterministic execution; and every result remains inside its task signature and evidence grade. This positioning accommodates the legacy CNN and teacher–student evidence as ablations while leaving static-MAP, Window-MAP, V5 early-stop, NMF-reproduction, auxiliary-CPD, and physical-board boundaries visible. It is neither a claim of a first adaptive GKP decoder nor a global comparison across CI, ML, direct NN, RL, AQEC, CPD, and FPGA systems.

4 Contract-centric dual-loop method

4.1 Method-state convention

This section uses four explicit method states. *Implemented and evaluated* denotes code that produced the frozen project results; *diagnostic only* denotes executed analysis that was not eligible for a confirmatory performance claim; *conditionally registered and stopped* denotes a task-board design whose continuation depended on a prior entry gate and whose implementation was never started; and *future physical work* denotes procedures that require the unavailable board or quantum apparatus. These states are methodological attributes, not missing-value conventions.

4.2 Protocol-aligned simulation stack

The implemented simulator is a single-mode square-lattice GKP abstraction, not a calibrated cavity–transmon digital twin. A synthetic drift process supplies continuous displacement mean, covariance, loss, and outlier variables together with discrete regime and burst state. The finite-squeezing layer separates channel, data-GKP, ancilla/measurement, and envelope contributions. The protocol layer adds stage-resolved ancilla faults, readout confusion, conditional reset, leakage persistence, and sharpen–trim or small–big–small state-machine bookkeeping. These components generate a causal modular syndrome stream and a logical label under the same registered lattice and correction convention.

Each simulated cycle produces two physically distinct records. The observed record contains the quantized syndrome, readout/event health, reset status, parameter age, and integrity fields that a deployable decoder may receive. A read-only truth record contains latent drift state, regime, unwrapped displacement, and logical outcome. Truth is used only to generate the channel, score errors, define evaluation-only event intervals, and construct the isolated oracle. It is absent from every online adapter, posterior, expert, bank image, and action. Prefix and truth-mutation tests enforce this separation. The vectorized engine supports trajectory-clustered Monte Carlo and rare-event strata, but host throughput is not interpreted as hardware latency.

4.3 System overview and reuse of the dual-loop pipeline

Figure 1 is the evidence-bounded system overview frozen by T7.1.2. It separates the per-round digital path, host update path, evidence ladder, timing owner, and physical-board null state on the same page.

The fast loop consumes the current validated bank on every measurement and never invokes the offline teacher. The slow loop may host analytical, filtering, or learned estimators, but each candidate must emit the same bounded image and pass the same admission, integrity, and rollback rules. MAP owns logical decoding, typed event/fallback logic owns safety authority, and the fixed-latency path owns deterministic action delivery. The diagram therefore acts as an interface map, not as evidence that the four-state student is the main LER winner or that a complete FPGA deployment has been measured.

4.4 Task formulation

At cycle t, the deployable decoder receives only quantized observed syndrome packets $o_{1:t}$ and health/event fields. It returns a logical action, frame update, fallback reason, active-bank identity, and versioned provenance. Truth fields used to generate trajectories are denied to all deployable adapters and are available only to the isolated evaluator and model oracle.

For each completed parameter window, the host side maintains two causal expert estimates:

$$\widehat{\theta}_t^{\mathrm{W}} = \mathrm{Window}(o_{1:t}), \qquad \widehat{\theta}_t^{\mathrm{E}} = \mathrm{EWMA}(o_{1:t}), \tag{1}$$

Each estimate compiles to a complete parameter-bank image for the same fixed-point fast path. The router may promote the Window bank only when the causal smooth posterior, pre-update predictive score, and hysteresis checks all pass. Tail or uncertain evidence uses the continuously updated and validated EWMA shadow; integrity faults alone may republish the last-known-good image.

4.5 Unified execution contract

A/B transactions write a complete inactive image, verify wire/image CRC plus manifest SHA, compare-and-swap against the active version, and activate only at a safe boundary. The retired bank remains protected

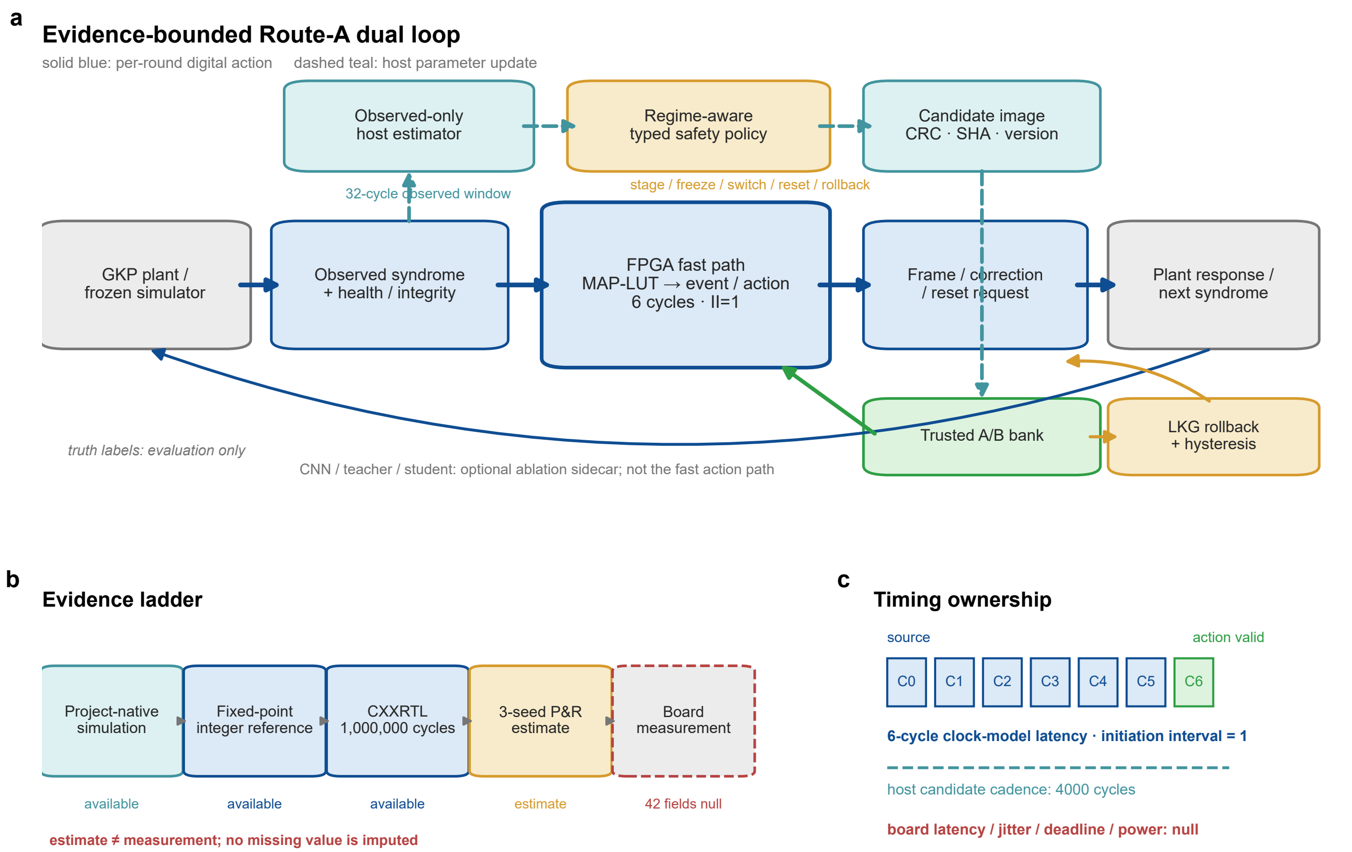


Figure 1: Evidence-bounded dual-loop Route-A contract. (a) Observed syndrome and integrity inputs feed a deterministic MAP-LUT/event fast path, while a software slow loop stages versioned parameter images through trusted A/B banks and last-known-good recovery. Solid and dashed arrows denote per-round and update-cadence paths, respectively; CNN/teacher/student modules remain optional ablations. (b) Available evidence progresses from project-native simulation through fixed-point and CXXRTL qualification to a three-seed post-route estimate; physical-board measurement is blocked and all 42 measured fields remain null. (c) The pre-board timing contract is six cycles with initiation interval one, whereas host updates occur every 4000 cycles. These values are not board-measured latency. Source data are provided in the T7.1.2 Source Data file.

Table 4: Frozen execution fields shared by deployable methods.

Field	Frozen value	Interpretation
Syndrome quantization	10-bit ADC; 8-bit LUT address; 2-bit fraction	Same quantized packet domain for every deployable comparator
MAP-LUT word	Signed Q9.12, ties-to-even, saturation	Deterministic arithmetic and packed-word behavior
Fast path	5-cycle MAP plus 1-cycle event/action; II=1	Six-cycle action-valid contract before transport
Bank protocol	CRC/SHA/CAS A/B bank, versioned commit, six-cycle drain	No partial publication; integrity-only monotonic LKG republish
Cadence	32-cycle posterior update; 2,048 scalar samples; 4,000-cycle image opportunity	Completed-window causality; equivalently 1,024 paired decisions and a 2,000-decision activation period
Matched ceilings	8,192 update MACs; 8,192 B state; 8,192 B workspace; 5,000 μs host update	Admission cap, while actual cost and every deadline outlier remain reported
Oracle isolation	Separate exact-key truth packet	Truth may score or bound performance but cannot select an online action

for six cycles. Host ack does not complete the transaction until bank, version, activation epoch, CRC, and SHA readback agree. Board-measured deadline miss is required to remain `null` before a physical run. The current hardware LUT is phase-conditioned and is not mathematically equivalent to the full two-dimensional joint MAP used as a strong software comparator. The latter is therefore retained in the comparison table but is not relabelled as current-RTL deployable.

4.6 Safety state and atomic updates

The implemented V4 slow loop uses a four-state causal HMM with class order {normal, smooth, calibration shift, burst}. Circular syndrome statistics and an observed heavy-tail event head are updated every 32 decisions. Model regularization, transition smoothing, and temperature were fixed to 0.1, 4.0, and 2.0 on calibration/pilot data. This model is an implemented synthetic-data classifier; it is neither an IMM nor BOCPD and it does not use an activation-horizon posterior predictor.

The six common policy thresholds were chosen from 1,728 pilot tuples; the selected tuple was $(0.9, 0.2, 0.25, 192, 2, 8)$ for enter probability, exit probability, uncertainty, OOD code, enter hysteresis, and recovery hysteresis. Window may be promoted for the next activation period only when the policy is open, the causal smooth posterior is at least 0.30, and its previous complete window has lower pre-update predictive negative log likelihood than EWMA. Tail/uncertain evidence permits only the validated EWMA shadow; leakage requests reset and blocks commits; CRC, SHA, version, age, deadline, or router-hash failure cancels algorithmic candidates and permits only monotonic LKG republishing.

The V4 policy replaced two failed architectures. A static-switch design and a freeze-all-EWMA design both failed all 38 pilot-safe candidate tuples because the supposedly trusted state became stale under persistent calibration or telegraph changes. V4 instead updates Window and EWMA shadow banks continuously. A 20,061-cycle structural trace produced versions 1–4 at cycles 4001, 8002, 12003, and 16005; all source-to-action fields were six cycles. At the tail transition, the EWMA shadow was committed rather than reverting to static. An integrity fault cancelled a pending Window update and completed an LKG rollback. The registered dual-shadow update cost was 1,218 MACs under the 8,192-MAC ceiling.

This prior-informed V1–V4 revision history is part of the method; the formal evaluation was result-blind only after V4 was locked, not from the beginning of the project. The structural trace proves execution and provenance, not tail-performance superiority. The independent T6.7.2 matrix subsequently established bounded non-inferiority to locked EWMA, as reported below.

The earlier presentation schematic below is retained because its interface structure survives the move

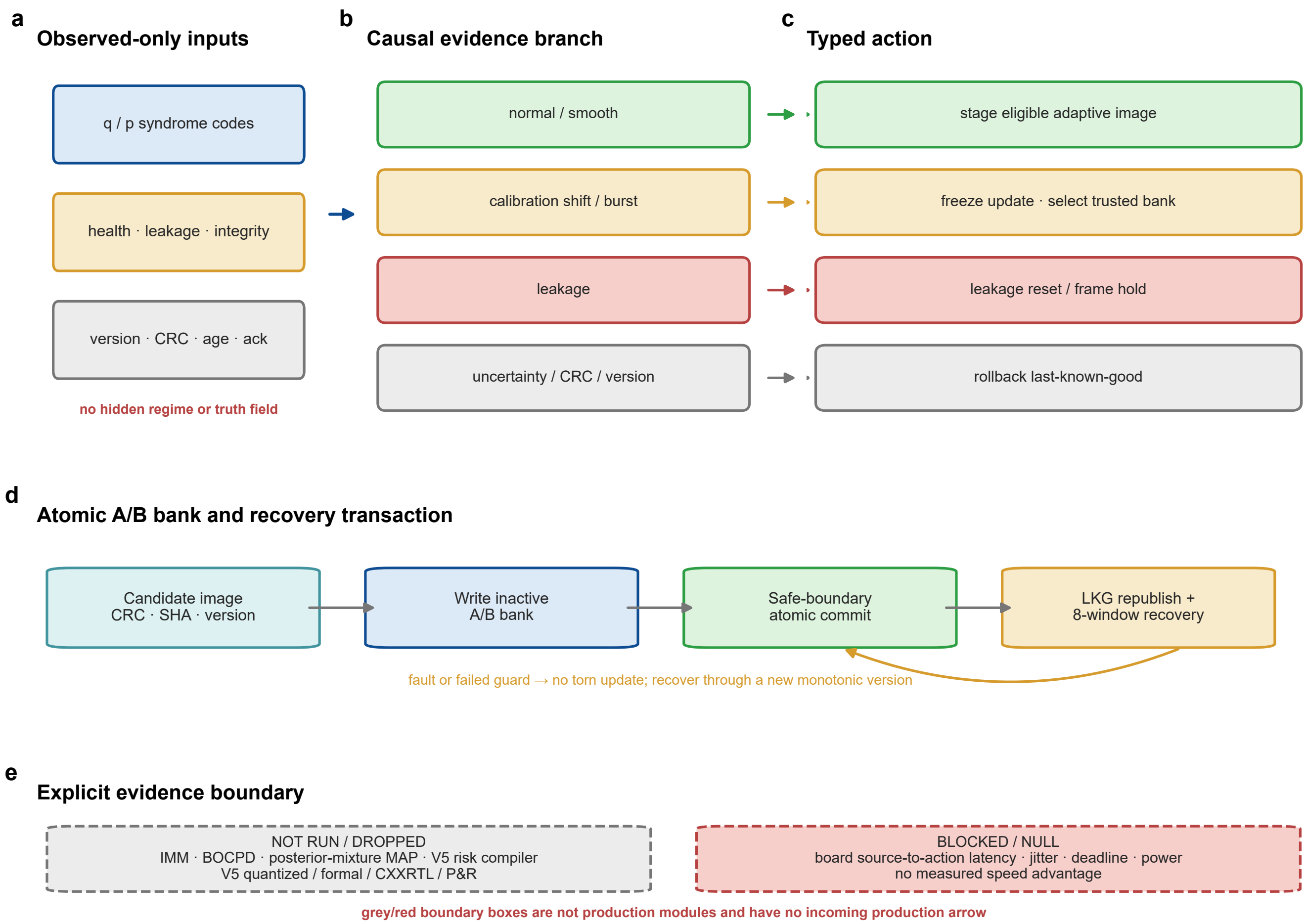
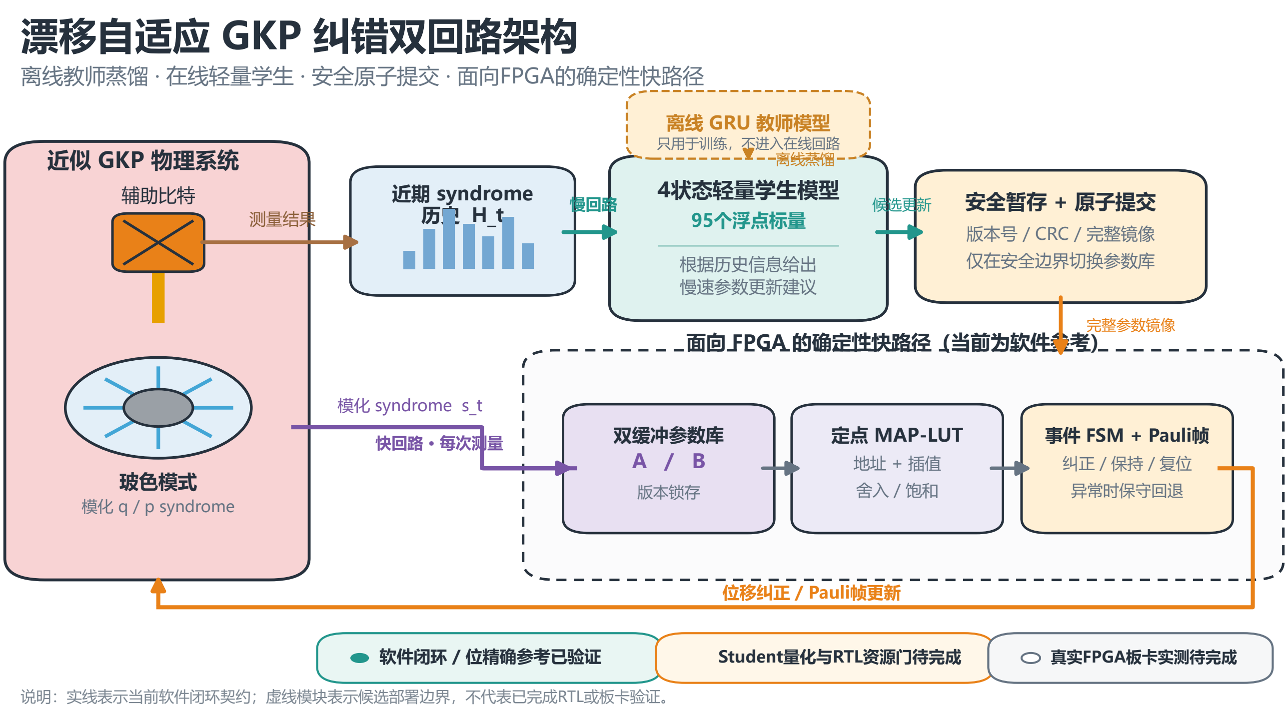


Figure 2: Typed fail-closed adaptation and atomic parameter control. (a–c) Observed syndrome, health, integrity, version, and age fields select typed normal/smooth, tail, leakage, and integrity branches and their corresponding stage, trusted-bank, reset, or last-known-good actions. (d) Candidate images cross CRC/SHA/version checks before inactive-bank write and safe-boundary atomic commit; recovery requires eight-window hysteresis. (e) IMM/BOCPD, posterior-mixture MAP, the V5 risk compiler, and all V5 implementation results were not run after the preregistered early stop, while board latency, jitter, deadline, and power remain blocked. Source data are provided in the T7.1.2 Source Data file.

away from a CNN-centric claim. It is presented as an unnumbered historical design aid so that the evidence-frozen main figures retain their registered numbering. The durable elements are observed syndrome history, a bounded slow-loop proposal, safe publication of a complete parameter image, versioned A/B banks, a quantized MAP-LUT, an event state machine, and a Pauli frame update. These interfaces, rather than the identity of one learned estimator, define the reusable architecture.



Historical presentation schematic | Reused Chinese overview of the drift-adaptive GKP dual loop. Under the current Route-A interpretation, the upper four-state student is an optional learning extension; Window and EWMA adaptive MAP provide the primary deployable parameter images, while HMM/event/fallback logic controls safe routing and publication. The maturity badges in this 16 July 2026 presentation asset are an archival snapshot: subsequent tasks added student fixed-point RTL and pre-board resource evidence, but real-board validation remains pending.

4.7 Matched baselines, oracle, and replaceable learned modules

The common-trace runner executes standard binning, static joint MAP, Window MAP, EWMA adaptive MAP, Kalman adaptive MAP, and V4 Route-A. Standard binning is compiled to the common LUT grid and checked exhaustively rather than bypassing the fast path. Static joint MAP remains a software-only strong baseline because full two-dimensional likelihood integration is not equivalent to the current phase LUT. The hidden-state oracle knows the simulator state and is reported only as an upper bound; it has no matched hardware or cost entry.

Learning is outside the critical claim path. A legacy CNN can propose a bounded update, a non-Markovian teacher can supply an offline control target, and a low-dimensional student can compress that target. All three must obey the same schema, cadence, bank, and failure rules. In the unified runner, the legacy CNN required 3,489,984 MACs, 1,586,368 B of state, and at least 65,536 B of workspace, so it was correctly demoted to ablation-only under the frozen budget. The offline non-Markovian teacher may use simulator truth to generate training targets, but no teacher output is an online action. Any distilled student must consume only observed history and pass the same packet, precision, cadence, bank, and deadline contract. This failure does not invalidate the contract; it demonstrates why learned modules must be replaceable rather than definitional.

4.8 Evaluation protocol and metrics

4.8.1 V4 result-blind formal design

The V4 formal protocol was frozen after disclosure of earlier counterexamples but before access to the new formal split. Calibration, pilot, and formal data use 12, 12, and 24 independent seed clusters; V4 did not use the later proposed V5 training partition. Calibration fit the HMM/event model, and pilot alone selected the baseline, architecture revision, and one common threshold tuple. The complete protocol contains 143 cells and 71,958,528 formal decisions per method across smooth and abrupt/OOD families. T6.7.1 consumed the smooth subset: four families, six held-out cells per family, 24 formal clusters, and 576 trajectories. T6.7.2 then consumed the locked abrupt/OOD and nominal subset: six fault families, one nominal cell, 24 formal clusters, and 888 trajectories. Together the two tasks exhaust the preregistered formal decision count without retuning the V4 lock.

All methods receive the same quantized trace. The primary comparator was selected once on pilot data and frozen as EWMA. Formal results cannot trigger baseline reselection, per-family threshold changes, family deletion, or router retuning. Inference uses 20,000 paired bootstrap replicates with the formal seed as the independent cluster and equal family weighting. Family-specific discoveries use Holm correction.

4.8.2 Metrics and offline scoring

The Pauli-resolved logical error rate is

$$p_{\mathrm{L}} = p_X + p_Y + p_Z = 1 - p_I. \tag{2}$$

The registered primary smooth contrast is

$$\Delta_{\mathrm{EWMA}} = p_{\mathrm{L}}(\mathrm{EWMA}) - p_{\mathrm{L}}(\textsc{Route}\text{-A}), \tag{3}$$

so a positive value favours Route-A. The static-to-oracle gap closure is

$$G_{\mathrm{static}} = \frac{p_{\mathrm{L}}(\mathrm{static}) - p_{\mathrm{L}}(\textsc{Route}\text{-A})}{p_{\mathrm{L}}(\mathrm{static}) - p_{\mathrm{L}}(\mathrm{oracle})}. \tag{4}$$

Negative G_{static} means that Route-A is worse than static MAP; it is not clipped to zero. Tail readouts use non-overlapping 512-decision windows, reporting p95 and global worst-window LER. Action diagnostics report fallback, unnecessary fallback, avoided and induced errors, false updates, commits, detection/recovery lag, and censoring separately from average LER. Simulator truth supplies only these offline scores and event intervals; it never enters a deployable feature or bank decision.

4.8.3 Evidence grades and stopping rules

Results are assigned to one of five non-interchangeable layers: literature fact, official-code reproduction, project-native matched simulation, fixed-point/CXXRTL/P&R pre-board evidence, or measured board/HIL evidence. Missing task adapters, source artifacts, checkpoints, or timing boundaries are reported as `N/A`, `null`, `blocked`, or `ineligible`; they are not imputed from a neighbouring lane. In particular, a positive Phase 6C decoder result cannot upgrade the V5 system gate, and a P&R estimate cannot populate a board-measured field.

4.9 Executed V5 entry diagnostic and conditional stopping branch

V5 was governed by an entry gate before estimator or RTL implementation. The already-opened V4 formal trajectories were used only to audit expert disagreement and a truth-privileged upper bound. Selection feasibility was then tested on a separate, non-formal development set of 186 trajectories and 4,571,136 decisions using leave-one-seed-cluster-out nested evaluation. The strict-causal selector could use current Window sufficient state, causal expert state, and the previous activation period's hard disagreement, but not family, cell, truth, or future observations. Cost was charged under the same 8,192-MAC/B limits.

Let H_{nested} be improvement of that deployable selector over its fold-selected baseline, and let H_{action} be the additional gain from expanding the correction-action family beyond the hard-expert decision oracle. Continuation required positive nested evidence and

$$H_{\text{action}} \geq 12\%. \tag{5}$$

Instead, $H_{\text{nested}} = -0.2322\%$, a held-out fixed mixture gave only $+0.4587\%$, and the pure mixture/action-space increment was nine avoided errors, or 0.02549% baseline-relative. The much larger truth-switched hard-expert oracle gap was not counted as attainable action-space gain. The entry verdict was therefore `NO_GO_V5_INSUFFICIENT_ACTION_SPACE_HEADROOM`.

The V5 items in Table 5 describe the stopped design, not implemented modules. The task board had conditionally registered a fresh train/calibration/pilot/formal four-way split, multiscale wrapped features, continuous-state static/trend/harmonic IMM, BOCPD/telegraph and robust-fault posterior, activation-horizon prediction, uncertainty-marginalized posterior-predictive MAP compilation, calibrated LER/CVaR gating, and typed two-bank actions. Because the entry gate failed, no four-split manifest, checkpoint, compiler image, V5 threshold, or production transaction was created.

Table 5: Evidence state of the conditionally registered V5 method.

Component	State	Consequence
Causal/action headroom audit	Diagnostic only; executed	Negative entry result may be reported, but not as a V5 performance experiment
Fresh four-way split and statistics lock	Conditionally registered; not instantiated	No V5 formal manifest, power calculation, or untouched split exists
Multiscale wrapped features; IMM/BOCPD; activation prediction	Dropped before implementation	V4 four-state HMM must not be renamed as any of these components
Posterior-predictive MAP and LER/CVaR risk calibration	Dropped before implementation	No posterior mean plug-in or Python mixture is promoted as a production V5 compiler
Typed expert/bank policy	Dropped before implementation	Existing V4 Window/EWMA/LKG transactions are not evidence for V5 bank residency
V5 fixed-point golden and long CXXRTL	Not run	V4 integer/CXXRTL traces cannot be relabelled as V5 retention or equivalence
Actual-module formal properties and multi-seed P&R	Not run	There is no V5 SVA/SMT atomicity proof, netlist, resource, Fmax, or power estimate
Physical-board measurement	Future work; blocked	Latency, jitter, deadline miss, power, and closed-loop fields remain null

4.10 Fixed-point, RTL, formal, and post-route evidence boundary

The implemented V4/pre-board path has an independent packed-word integer reference, complete fixed-point saturation and ties-to-even rules, and CXXRTL cycle comparison. The integrated qualification replayed ten 100,000-cycle families, including frozen observed trajectories and directed threshold, commit-race, version, CRC, leakage, reset, deadline, and FIFO faults. All visible action/state words matched and no undefined action or silent overflow was observed. This is sampled long-sequence equivalence and failure-path coverage, not exhaustive formal verification.

An open-source synthesis/place-and-route flow evaluated the parameterized V4 top with and without the optional student at seeds 1, 7, and 19. It binds netlists, reports, logs, memories, and source hashes, but reports an estimate, not vendor signoff or board timing. The planned actual-parameterized V5 SVA/SMT proof, V5 long CXXRTL, and V5 P&R were all stopped. Consequently the Methods include formal verification as an explicit absent branch rather than conflating the V4 mutation/CXXRTL evidence with a proof over all bank states.

4.11 Future physical-board procedure

The available UART/bitstream package is a pre-board candidate only. A physical run must first bind the actual board and revision, pinout, transport, timestamp method, bitstream, and source hashes. It must then repeat at least one million cycles with a streaming or autonomous trace so that serial request gaps do not replace II=1 input semantics, and measure core, transport, source-to-action, and end-to-end latency separately. Until that procedure is executed, all 42 board-measured fields remain null and no measured speed, zero-deadline-miss, power, or faster-than-existing-FPGA claim is admissible.

5 Results

5.1 Result-state and ranking convention

Results are ordered by evidential role rather than by effect size. We first report the complete V4 system comparison, including comparator wins, family concentration, operational costs, and hardware-measurement nulls. We then report the executed causal-headroom diagnostic and the resulting V5 early stop. Only after that terminal decision do we show Phase 6C results, which use independent secondary splits and cannot alter the V4 or V5 verdicts. Historical CNN and teacher–student results follow as replaceable extension evidence.

Table 6: Result-state contract used throughout this section. A row's position does not create cross-task comparability.

State	Admissible content	Ranking rule
Primary restricted	Frozen V4 paired simulation and pre-board execution	Rank only methods sharing the V4 packet, split, observation, action, and budget contract
Mandatory negative	Static/Window wins, tail equality, fallback/false-update cost, failed families and attempts	Report beside the positive contrast; never offset with another lane
Diagnostic stop	Executed V5 causal/action-space entry audit and downstream absence	Report the stop decision, not a V5 decoder-performance result
Secondary eligible	Official-code reproduction or project-native matched Phase 6C cells	Compare only within the frozen task signature; label every result secondary
Null or contextual	blocked, ineligible, or `LITERATURE_ONLY` cells	Preserve as a boundary ledger or Supplement/Related Work; no numerical ranking

This convention also fixes the denominator problem. A positive multimode CPD result cannot rescue the stopped single-mode V5 route; a literature latency cannot replace a board measurement; and an oracle or unavailable-truth result cannot enter a deployable ranking. Counts of failed families, restarts, attempts, and missing outputs are retained because their absence would change the scientific interpretation even when no scalar metric is available.

5.2 Untouched smooth formal comparison

Table 7 reports the complete T6.7.1 method ordering. It contains the result that controls the current LER claim.

Table 7: T6.7.1 smooth formal result. Each method has 28,311,552 scored decisions. The oracle is nondeployable. Window statistics use 512 decisions.

Method	Aggregate p_L	p95 window LER	Worst window LER
Standard binning	9.7861×10^{-4}	3/512	17/512
Static joint MAP	9.6819×10^{-4}	3/512	16/512
Window MAP	8.9642×10^{-4}	2/512	64/512
EWMA adaptive MAP	1.01443×10^{-3}	2/512	59/512
Kalman adaptive MAP	1.78574×10^{-3}	3/512	85/512
Route-A	9.92740×10^{-4}	2/512	59/512
Hidden-state oracle	1.62337×10^{-4}	1/512	7/512

The primary contrast passed:

$$\Delta_{\mathrm{EWMA}} = 2.1687 \times 10^{-5}, \qquad \mathrm{CI}_{95\%} = [1.9003, 2.4548] \times 10^{-5}. \tag{6}$$

This is a 2.14% relative reduction from the locked EWMA LER. The gain was not uniform across smooth families.

Table 8: Family-specific EWMA-minus-Route-A contrasts.

Family	Difference	Paired 95% interval	Holm discovery
Mean drift	5.65×10^{-7}	$[-2.83 \times 10^{-7}, 1.41 \times 10^{-6}]$	No
Variance drift	0	$[-7.06 \times 10^{-7}, 7.06 \times 10^{-7}]$	No
Correlation drift	0	$[0, 0]$	No
Periodic drift	8.618×10^{-5}	$[7.545, 9.735] \times 10^{-5}$	Yes

Against stronger descriptive comparators, the conclusion reverses. Route-A is worse than static joint MAP by 2.45×10^{-5} and worse than Window MAP by 9.63×10^{-5}. The static-to-oracle gap closure is

$$G_{\mathrm{static}} = -0.03046, \qquad \mathrm{CI}_{95\%} = [-0.04966, -0.01119]. \tag{7}$$

Thus the current smooth result supports a pre-registered EWMA contrast and a periodic-drift effect. It does not support "best deployable decoder", "better than static GKP decoding", or "general smooth-drift advantage".

A pinned Bayesian online change-detection wrapper was also replayed on the common traces. Its paired LER exceeded Route-A by 1.00184×10^{-5} with a 95% interval $[7.66793 \times 10^{-6}, 1.23689 \times 10^{-5}]$. This is not evidence for a general drift-adaptive SOTA claim: the wrapper's worst update took 13,004.1 μs against the frozen 5,000-μs cap and incurred one deadline miss. The outcome and the matched-budget failure are therefore reported together, and the method is excluded from the strongest-deployable-baseline denominator rather than being counted as a favourable external win.

5.3 Action and update diagnostics

The formal action ledger contained 884,736 scored posterior updates and 13,824 commits. Of those commits, 2,870 used the Window bank and 10,954 used the EWMA bank. There were 661 avoided errors and 47 induced errors relative to EWMA, with zero false updates under the smooth definition. However, the fallback signal rate was 21.6305%, and the unnecessary-fallback decision rate was 21.5822%. Every trajectory had an adaptation lag of 1,904 decisions.

The net avoided-minus-induced count is encouraging for the locked smooth contrast, but the high fallback rate shows why average LER cannot substitute for a tail-safety experiment. Earlier T5.4.2 evidence reinforced this caution: generic uncertainty gating reduced aggregate OOD error by 0.00107490 with a 95% interval $[0.00001950, 0.00227615]$, yet it significantly harmed the compound family by -0.00488536 and imposed a small nominal cost. T6.7.2 therefore reported every abrupt/OOD family and the nominal negative control without using the periodic smooth gain to offset a failure.

Policy selection also retained two complete failed families. The static-switch and freeze-all-EWMA architectures each failed all 38 posterior-safe pilot tuples; their closest minimum safety slacks were -0.037109375 and -0.044921875, respectively. No tuple from either family was promoted, and their failure is part of the rationale for the V4 dual-shadow design rather than an unreported hyperparameter restart.

5.4 Abrupt/OOD and nominal non-inferiority

The independent T6.7.2 matrix contains 888 trajectories and 43,646,976 scored decisions per method across step, telegraph, burst, readout/reset, leakage, compound, and nominal conditions. All registered catastrophic-degradation, calibration-strict, and nominal non-inferiority gates passed. The result must nevertheless be read as a safety bound against the locked EWMA baseline rather than as a decoder-performance gain: step, telegraph, readout/reset, leakage, and compound had exactly zero Route-A–EWMA difference in all core safety statistics. Burst was aggregate-neutral, with a maximum single-window excess of one error and two avoided versus two induced errors.

Table 9: T6.7.2 calibration-shift and nominal results. Equality to EWMA passes the registered non-inferiority gates but is not an advantage over that comparator. Window statistics contain 512 decisions.

Method or diagnostic	Average LER / rate	Global worst / bound
Static joint MAP, calibration shift	0.01087	32/512
Window MAP, calibration shift	0.01243	182/512
Locked EWMA, calibration shift	0.01227	181/512
Route-A, calibration shift	0.01227	181/512
Nominal fallback	0.11936%	registered bound 1.0%
Nominal unnecessary fallback	0.11783%	registered bound 0.75%

The calibration result removes the earlier degradation only relative to locked EWMA: both methods have a worst window of 181/512 and a seed-worst paired difference of $0\,[0, 0]$. Static MAP is substantially better at 32/512 and lower average LER. Nominal Route-A–EWMA average difference and induced-minus-avoided count are also $0\,[0, 0]$.

The operational cost is large in the tail cells. Fallback signals span 59.45–95.85%, unnecessary fallback spans 58.80–94.63%, and 2,044–3,365 of 3,456 commits per family occur inside the pre-registered tail truth interval and are therefore counted as false updates. V4 blocks Window promotion under tail evidence but continues to update and publish the trusted EWMA shadow; it must not be described as freeze-all adaptation. Detection had no censored events (burst p95 96 decisions), and recovery-to-OPEN also had no censoring in the recoverable families (p95 256–320 decisions). These diagnostics support auditable behavior and registered non-inferiority, while directly ruling out claims of lower tail LER, low fallback, or superiority to static MAP.

5.5 Deterministic execution and pre-board hardware evidence

The latency advantage currently established is determinism and feasibility, not measured superiority over another decoder.

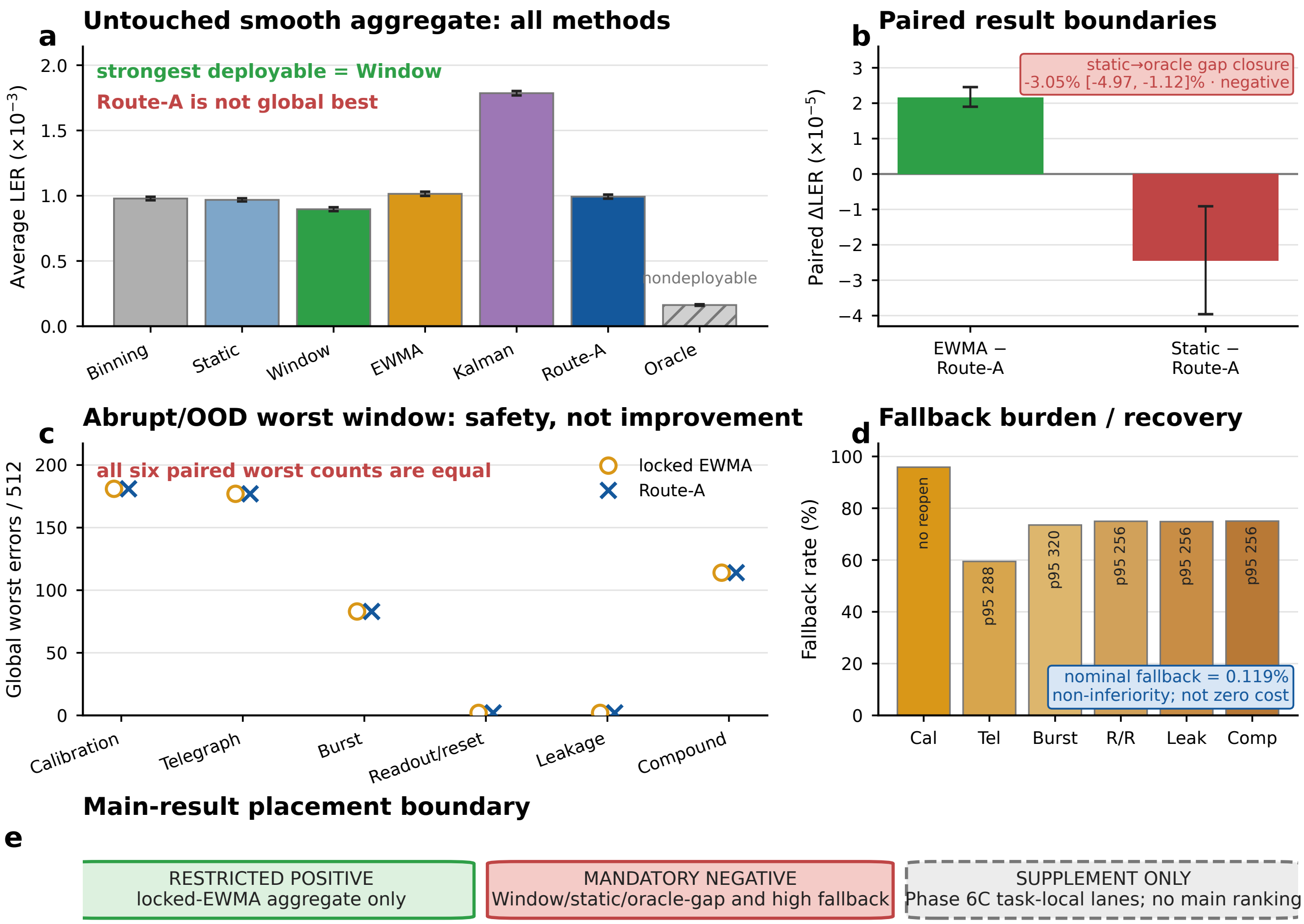


Figure 3: Restricted V4 performance and safety results. (a) Smooth results use 24 formal seed clusters and equal family weights. ROUTE-A improves only over the pilot-locked EWMA aggregate; Window remains the strongest deployable method and the oracle is nondeployable. (b) The paired static contrast and static-to-oracle gap closure are negative. (c) Across six abrupt/OOD families, paired worst-window outcomes establish safety and non-inferiority rather than improvement. (d) That boundary requires substantial fallback and recovery cost. (e) Positive Phase 6C results are excluded from the main ranking and retained only in task-local Supplement lanes. Source data are provided in the 55-row T7.1.3 Source Data file.

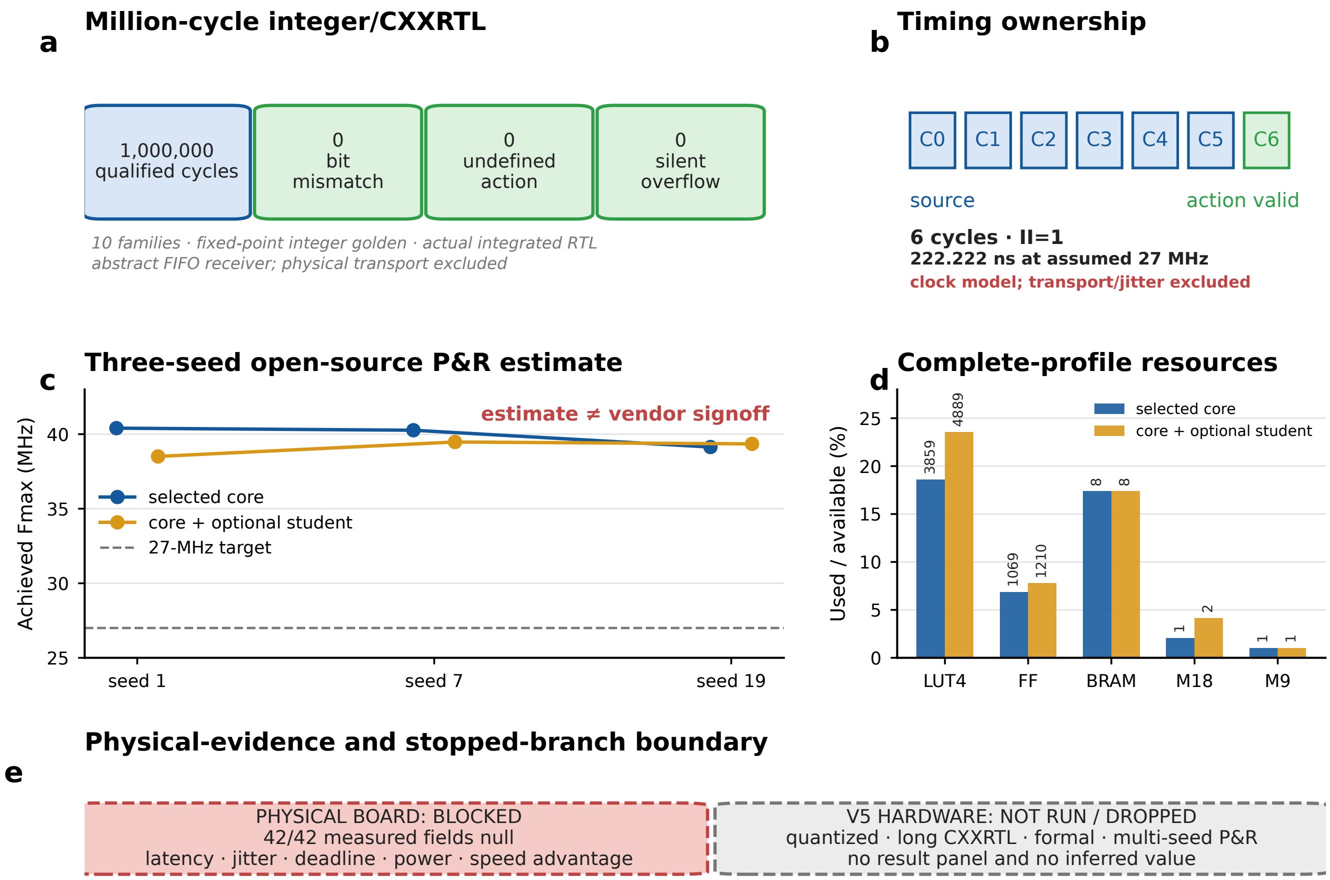


Figure 4: Pre-board deterministic execution evidence and physical-measurement boundary. (a) The integer/CXXRTL path completed 1,000,000 cycles with zero bit mismatch, undefined action, or silent overflow. (b) The integrated selected profile has a six-cycle, initiation-interval-one clock-model path; 222.222 ns is only the conversion at the assumed 27-MHz contract clock. (c) Both the selected core and the optional-student sidecar cleared 27 MHz in three open-source placement-and-routing seeds. (d) Resource values describe complete integrated profiles rather than a MAP-ROM-only core. (e) All 42 physical-board fields remain null, and V5 quantized, formal, CXXRTL, and P&R stages were not run after the preregistered early stop. Source data are provided in the 55-row T7.1.3 Source Data file.

Table 10: Hardware-facing evidence layers. P&R numbers are open-tool estimates, not vendor signoff or board measurements.

Surface	Quantitative result	Supported interpretation
Source-bound static-MAP core	Three-seed Fmax 41.024–42.212 MHz; at most 3,387 LUT4, 865 FF, 8 BRAM and 2 DSP; six cycles and initiation interval one	A compact current-source core clears the 27 MHz contract clock in post-route estimate
Integrated V4, no student	Minimum Fmax 39.137 MHz; at most 3,859 LUT4, 1,069 DFF and 8 BSRAM; six-cycle source-to-action path	The contract, router, bank and action path fit together in the selected open-tool target
Integrated V4 plus 4-state student	Minimum Fmax 38.506 MHz; at most 4,889 LUT4, 1,210 DFF and 8 BSRAM	A compressed replaceable student remains feasible outside the per-shot critical path
Integrated CXXRTL qualification	4 smooth plus 6 fault families, each 100,000 cycles; zero visible-field mismatch, undefined action, route/core CRC error, or silent overflow; 19/19 gates	Board-independent bit-accurate execution and recovery qualification

The integrated qualification also exercised six modes, five health states, three actions, nine reachable fault bits, CRC/version failures, FIFO disturbance, commit rejection, and counter/LLR saturation. The directed fault family retained all 75 host-commit attempts and 25 rollback attempts; none is removed from the denominator because an assertion or recovery branch was unfavourable. The HMM itself remains a software slow-loop component; the synthesized design consumes the quantized posterior/policy overlay. Formal replay was 99.5802% real traces and 0.4198% directed branch vectors. This evidence does not cover real transport, clock-domain crossing, metastability, bitstream generation, power, ADC/DAC latency, or board deadline distributions. No measured FPGA latency, end-to-end latency, or external fastest-decoder claim is active.

5.6 Causal-headroom audit and V5 early stop

The V4 NO-GO triggered a causal audit before any V5 model or RTL was built. Across 71,958,528 diagnostic expert decisions, the best deployable aggregate baseline was EWMA with $p_{\mathrm{L}} = 0.01154953$. Truth-privileged selectors appeared to offer substantial improvement—2.4737% at family level, 4.1784% at cell level, 16.665% when activation truth was exposed, and 29.523% per decision— but these values use information unavailable to an online decoder.

The deployable audit used 186 development trajectories and 4,571,136 decisions. A strict causal selector produced 35,396 errors against 35,314 for its nested baseline, or -0.2322% headroom. A separately frozen held-out mixture produced 35,152 errors, only 0.4587% better than the nested baseline. More decisively, the hard decision oracle had 29,105 errors and the expanded action oracle had 29,096: nine errors separated them, corresponding to only 0.02549% incremental action-space headroom versus the 12% entry gate.

The resulting verdict was `NO_GO_V5_INSUFFICIENT_ACTION_SPACE_HEADROOM`, later frozen as `NO_GO_V5_EARLY_HEADROOM_STOP`. Tasks T6.10.2–T6.15.4 were dropped. Consequently there is no V5 untouched LER matrix, tail test, quantized-retention result, formal proof, CXXRTL run, or P&R profile. Reusing V4 hardware artifacts as if they belonged to V5 would violate the evidence contract. The early stop is therefore a substantive negative result: the observed oracle gap primarily calls for better causal state estimation or a different physical task, not a larger correction-action menu.

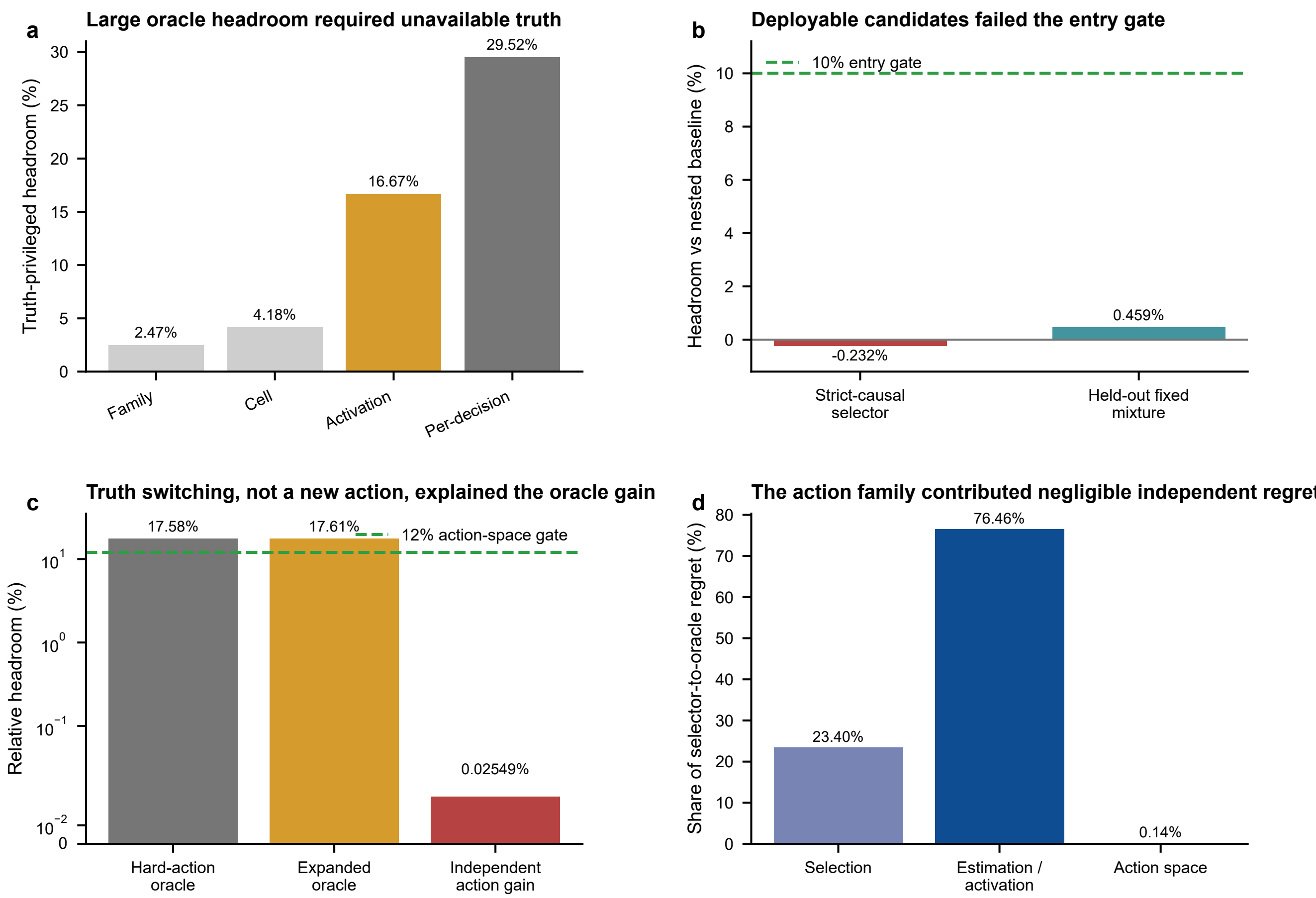


Figure 5: Why V5 was stopped before implementation. (a) Large diagnostic oracle headroom grows with access to unavailable truth. (b) Deployable selectors do not approach the registered 10% entry level. (c) Hard and expanded truth oracles are almost identical; the new action family contributes 0.02549%, far below the 12% action-space gate. The logarithmic axis retains the small positive value. (d) The selector-to-oracle regret is dominated by estimation/activation (76.46%), not action-space capacity (0.14%). Source: T6.10.1 causal-headroom JSON/CSV; percentages are independently recomputed by the plotting script.

5.7 Independent Phase 6C decoder and comparison lanes

After the V5 stop, Phase 6C was executed as a read-only auxiliary programme. Its split and hashes cannot enter the V5 denominator, and each result keeps its own task and evidence grade. This separation matters because the strongest new algorithmic result is positive even though the system-promotion result is not. The final atlas contains 206 cells and passed 24/24 integrity gates, 24/24 single-gate mutations, and 162 joint regressions. Passing means that provenance, metric recomputation, null/negative states, and evidence boundaries are intact; it does not mean that every lane has a positive result.

Table 11: Phase 6C rows eligible for explicitly secondary Results. Every row has evidence grade `OFFICIAL_CODE_REPRODUCTION` or `PROJECT_NATIVE_MATCHED`; rows are intentionally not combined into a global score.

Lane	Evidence grade	Result	Allowed interpretation
Single-mode square CI/CPD	Project-native matched	Exact separability plus 1,048,576-point q10 audit; zero CI–Euclidean-CPD hard-action mismatches	Geometric equivalence only; likelihood/coset MAP remains distinct
Two-GKP CNOT CI/ML	Project-native matched reproduction	3,080,192 trials; ML reduced gate failure by 31.160%, 58.065%, and 67.982% at 9, 12, and 13 dB; maximum literature discrepancy 1.52%	Analog ML helps for the reproduced matched gate model
Structured surface–GKP CPD	Official-code reproduction	2,005/2,005 upstream checks; official-data CPD/analog means 0.602456/0.599594; independent $d = 3, 5, 7$ crossing means 0.599298/0.600245; CPD lower LER in 27/27 cells	Official-data reanalysis plus partial small-distance reproduction, not full-distance threshold precision
Multimode posterior-weighted CPD	Project-native matched	9.6 million cycles, 38.4 million modal decodes; adaptive $p_{\mathrm{L}} = 0.172261$ versus static Euclidean 0.236929	Observed-only weighting gives a large drift gain in this independent model
AQEC wall-clock replay	Project-native matched	Six cells $\times$ 24 clusters on a common 700-μs horizon; measurement-feedback and autonomous control failed to beat idle in every cell	Negative finite-cutoff project result; not an official AQEC reproduction or refutation

The corresponding non-ranking boundary ledger is separate from Table 11. Learned eligibility was zero of 16 candidate families; the exact GQF source checklist passed zero of 15 items and left all 13 matched NMF metrics null; official AQEC protocol reproduction remained blocked; and 18 external FPGA implementations yielded zero exact same-task rows. These counts document failed or unavailable comparisons, not additional performance observations. Literature-only latency and lifetime values remain in Related Work or the Supplement and are never juxtaposed with project values as a numerical ranking.

For the multimode task, static posterior weighting alone was harmful in the aggregate ($p_{\mathrm{L}} = 0.261679$), whereas the observed-only adaptive rule nearly matched the metric oracle (0.172261 versus 0.171929). Its absolute improvement over static Euclidean decoding was

$$0.0646684, \qquad \mathrm{CI}_{95\%} = [0.0644128, 0.0649264], \tag{8}$$

and all 32 seed clusters favoured the adaptive lane. The worst-window LER fell from 0.320313 to 0.291016 and CVaR_{95} from 0.277271 to 0.240627. These are the clearest current LER and tail-metric advantages, but they belong to the independent multimode CPD task rather than to the stopped V5 policy.

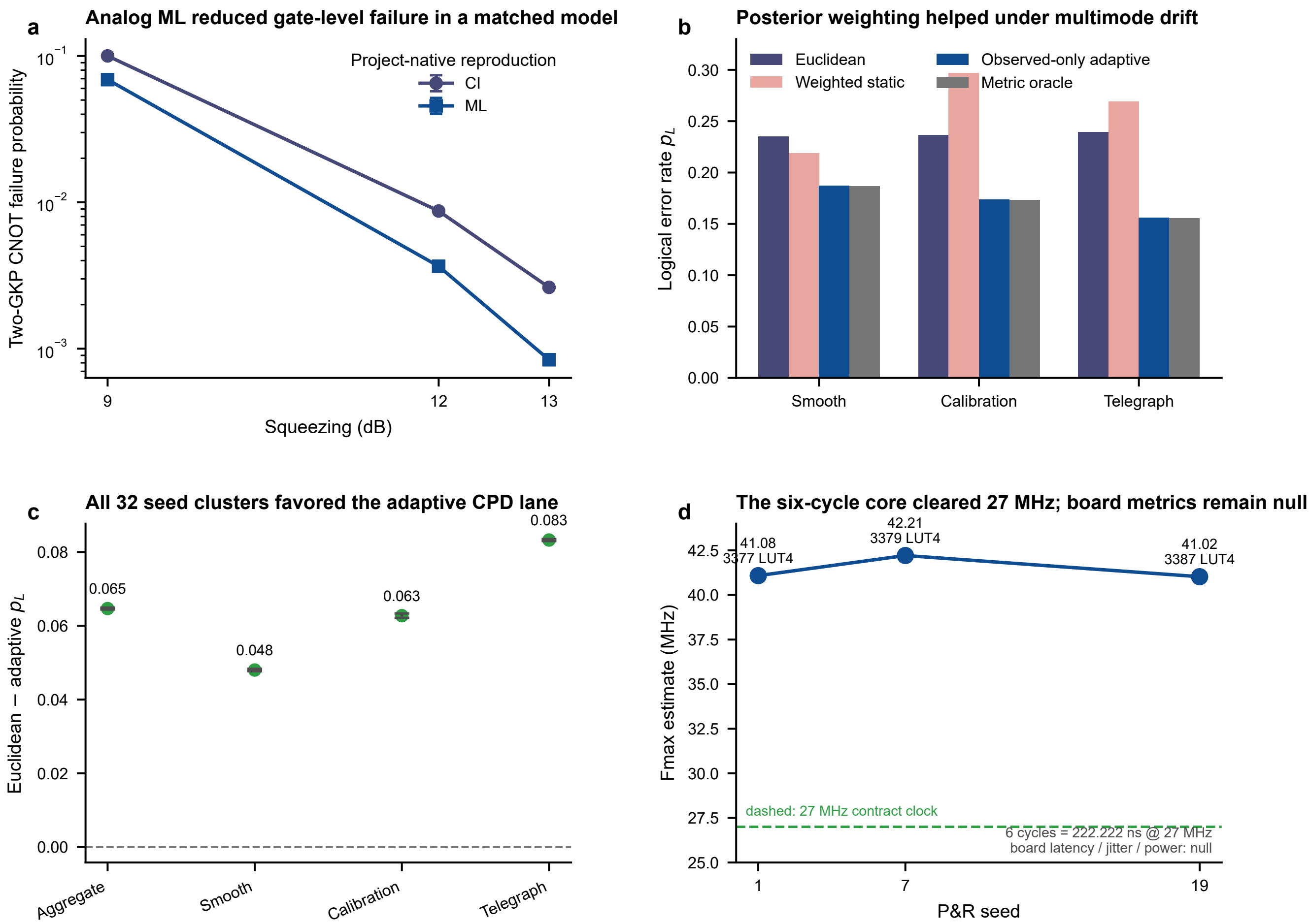


Figure 6: Bounded positive evidence outside the stopped V5 route. (a) Matched two-GKP-CNOT CI/ML reproduction; points show binomial estimates and 95% intervals. (b) Scenario LERs for the independent multimode CPD task. (c) Static-Euclidean-minus-adaptive LER is positive in the aggregate and each family; error bars are paired seed-cluster 95% intervals. (d) A source-bound six-cycle static-MAP core clears 27 MHz in three P&R seeds, but 222.222 ns is a contract-clock estimate and board latency, jitter, and power remain null. Sources: T6.17.2, T6.18.3, and T6.19.1 machine artifacts.

T7.1.4 retains the six-lane atlas as Supplementary Figure S5, alongside four new evidence-bounded supplementary figures. Moving the atlas out of the main result sequence is a placement decision, not an evidence downgrade: it prevents a heterogeneous comparison map from being read as a fifth main-result ranking.

5.8 Legacy CNN+FPGA evidence retained as an extension lane

The earlier T24 frozen-set benchmark remains valid within its original mock-backed software-HIL scope. Hybrid Residual-B was the winner and UKF the runner-up in all four scenarios.

Two later scenario-specific repeat sets strengthened only the direction of this historical result. With 12 paired formal-length repeats, the mean UKF-minus- hybrid differences were 0.015563 for static bias (paired-t 95% interval $[0.014778, 0.016348]$) and 0.022417 for linear ramp ($[0.021215, 0.023619]$); all 12 pairs were positive in both scenarios. The other two scenarios, all-scenario pooled inference, held-out robustness, and

Table 12: Historical T24 result retained as learning-extension evidence. The relative reduction is $(p_L^{\mathrm{UKF}} - p_L^{\mathrm{hybrid}})/p_L^{\mathrm{UKF}}$. This table is not part of the new unified ROUTE-A ranking.

Scenario	UKF final LER	Hybrid final LER	Relative reduction
Static bias+θ	0.825370	0.810902	1.75%
Linear ramp	0.811201	0.787755	2.89%
Step σ+θ	0.811548	0.788800	2.80%
Periodic drift	0.821558	0.806392	1.85%

hardware validation were not closed by that evidence.

The proper reuse is therefore modest: a learned residual can be useful on a bounded affine parameter surface. It is not evidence that CNN is the best slow-loop estimator, that it satisfies the new matched budget, or that the complete system wins in tail and latency.

5.9 Teacher–student extension evidence

The non-Markovian teacher/student line remains useful as a replaceable control extension motivated by feedback-based QEC and real-time experimental control [28, 6]. Figure 7 reuses the presentation result panel because it exposes both comparator ordering and the paired-seed sample structure. The metric is a ten-cycle, finite-horizon area-equivalent effective fidelity lifetime; it is neither an asymptotic decay constant nor FPGA wall-clock time.

The student closely tracked the offline teacher: their cutoff-12 lifetimes were 8.427 and 8.440 cycles, and their cutoff-16 lifetimes were 9.489 and 9.525 cycles. The five-agent MF mean was slightly higher at cutoff 12 (8.623 cycles) and lower at cutoff 16 (9.156 cycles), so the figure does not support universal teacher superiority. Across the three preregistered retention metrics, the lowest student point retention was 0.981457 and the lowest 95% interval lower bound was 0.944501 relative to teacher-versus-standard gain.

Selection multiplicity is retained rather than collapsed to the displayed candidate. The fresh teacher used three restarts (601, 709, and 811) and was selected by validation only; restarts 601 and 811 hit the 320-epoch cap, and a test-hindsight restart would have differed but was not substituted. The student search used 1-, 2-, and 4-state models with three restarts each; all six 2-/4-state fits hit the 900-epoch cap, while evaluation remained excluded from restart and dimension selection. These cap hits and ranking reversals prevent an optimizer-optimality claim even though the frozen candidate passed its retention gate.

The selected student uses four recurrent states and 95 stored scalars, versus 72,853 for the teacher, a $\sim 767\times$ float-model compression. Its fixed-point RTL produced zero mismatches across 7,680 outputs, with maximum fixed-versus-float error 1.46038×10^{-4}.

This is evidence for gain retention and compression within the project model. It is not an official reproduction or improvement over Puviani *et al.*, not a general memory-mechanism result, and not the main decoder LER claim. The larger quantized GRU lower-bound required 72,854 cycles (2,698.30 μs at 27 MHz), exceeded the five-microsecond slot, and was dropped. That negative result supports the architectural decision to keep learning outside the critical fast path.

5.10 Supplementary validation and failure-domain audit

The supplementary contract contains 792 records across 17 categories. Each record carries a panel, metric, condition, value or interval, evidence state, source selector, and source hash. The contract passed 17/17 gates and substantive mutations, all 13 live parent verifiers, and 210 parent-chain tests split across the dependency-complete and frozen Phase 6C numerical environments. This suite adds audit depth, not a new system-promotion result.

Three quantitative patterns are nevertheless informative. First, the noise-transfer surrogate is explicitly regime dependent: its relative error against the direct/Fock calculation is 0.56037 at 3 dB, but falls to

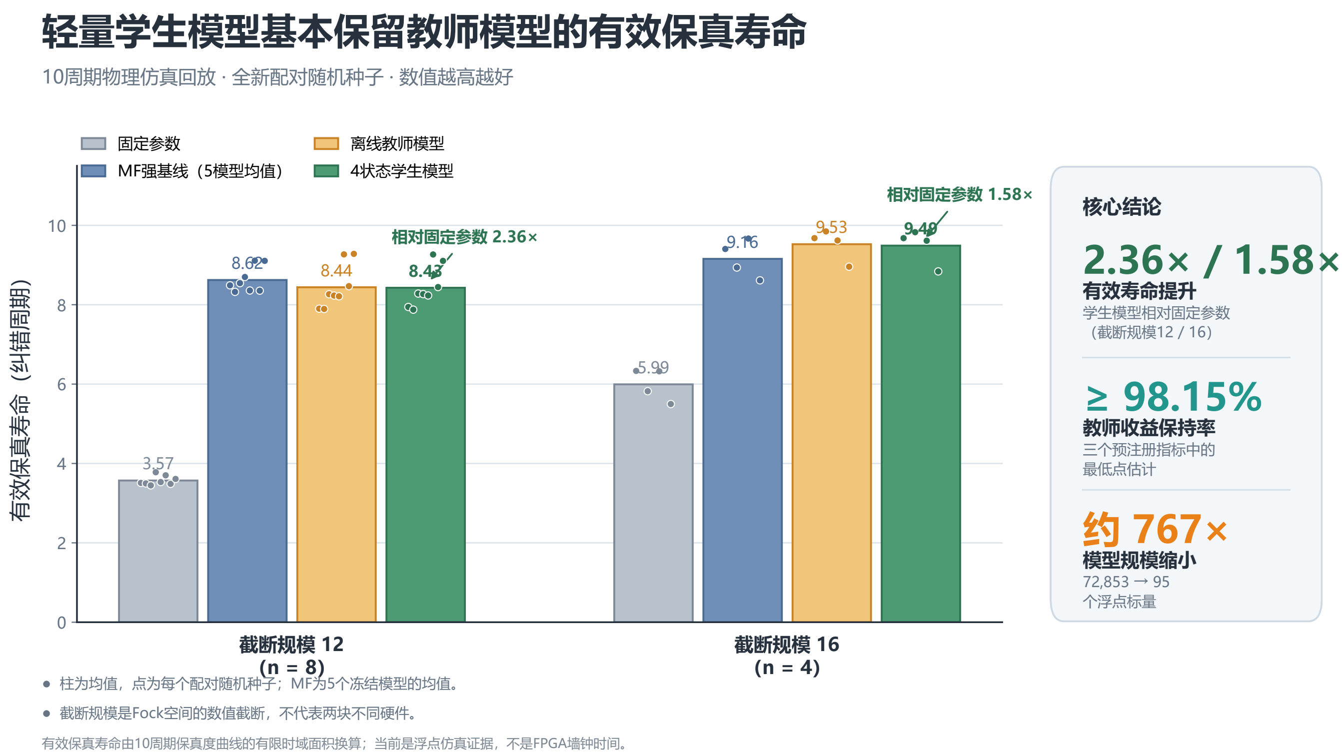


Figure 7: Finite-model teacher–student extension evidence. At cutoff 12 ($n = 8$ paired evaluation seeds), the four-state student reached an effective fidelity lifetime of 8.427 cycles versus 3.571 for fixed parameters, a 2.36× ratio. At cutoff 16 ($n = 4$), the corresponding values were 9.489 and 5.994 cycles, a 1.58× ratio. Bars show seed means and points show paired seeds; the MF bar is the mean of five frozen agents, not a test-selected winner. The sidebar's 98.15% is the lowest point retention, whereas the lowest paired-bootstrap 95% interval lower bound is 94.45%. All quantities are float finite-model results.

7.4845×10^{-3} at 10 dB and 1.3473×10^{-4} at 12 dB. The low- squeezing point is therefore a counterexample, not a calibration target. Second, all six finite-cutoff Pauli-eigenstate curves retain a QEC-on/QEC-off separation in the registered CPTNI simulation, but this is neither infinite-dimensional convergence nor experimental tomography. Third, the selected production integer profile gives quantized-minus-float LER 3.0518×10^{-5} with interval $[-4.5776 \times 10^{-5}, 1.2207 \times 10^{-4}]$, while the dense profile is bit-identical on the registered replay. These are representation-retention results, not measured FPGA advantages.

The same supplement makes the failure modes harder to omit. In the software communication abstraction, the compound jitter–burst case increases LER from 0.01986 to 0.06284 while control availability falls from 0.94899 to 0.08102. Supplementary Figure S4 also keeps nine terminal states visible: the low-squeezing failure domain, host memory/runtime boundaries, unsynthesized top-K, an incomparable Petz–student question, lane-local OOD, no broad tail gain, V5 Dropped, and 42 null board fields.

6 Where the current data show an advantage

Table 13 is the compact answer to which metrics currently show an advantage. Rows are deliberately not collapsed into a global score because decoder, controller, and hardware lanes have different objects and units.

Table 13: Metric-level advantage matrix at the T7.1.4 evidence cutoff.

Metric / property	Status	Evidence-supported statement	Boundary
V4 smooth LER vs locked EWMA	conditional	ROUTE-A reduced aggregate LER by 2.1687×10^{-5}, 2.14% relative, with a positive paired interval	Only the preregistered EWMA contrast; only periodic drift survived Holm correction
V4 LER vs static / Window	not established	ROUTE-A LER is higher than static and Window; static-to-oracle gap closure is negative	Direct counterevidence to a general V4 decoder advantage
V4 abrupt/OOD vs locked EWMA	conditional	All catastrophic, calibration-strict, and nominal non-inferiority gates passed	Five families are ties; burst is aggregate-neutral; this is not a tail advantage
V4 tail LER vs static MAP	not established	Calibration worst was 181/512 for ROUTE-A versus 32/512 for static MAP	High fallback and false-update rates remain visible
V5 deployable headroom	not established	Nested causal selector -0.2322%; held-out mixture $+0.4587\%$; new action family only 0.02549% versus 12% gate	V5 stopped before formal, quantized, RTL, or P&R work
Multimode CPD LER and tail	supported	Observed-only adaptive CPD reduced aggregate LER by 0.0646684 with positive 95% interval; all 32 clusters positive; worst-window and CVaR_{95} improved	Independent project-native multimode task; cannot upgrade V5
Two-GKP CNOT analog ML	supported	Failure reductions of 31.160%, 58.065%, 67.982% at 9, 12, 13 dB across 3,080,192 trials	Matched gate-level reproduction, not a memory or hardware result

Metric / property	Status	Evidence-supported statement	Boundary
Structured surface–GKP CPD	conditional	Official-code small-distance crossing reproduced; CPD lower LER in 27/27 cells	Partial distance range; no full threshold or project hardware claim
Fast-path determinism	supported	Six-cycle contract and one-million-cycle integrated CXXRTL run with zero visible mismatch; 19/19 gates	V4 pre-board result, not V5 or board measurement
Target-device feasibility	supported	Integrated V4 minimum Fmax 39.137 MHz; source-bound static core 41.024–42.212 MHz; bounded resources	Open-tool estimate; 222.222 ns is a contract-clock estimate
Teacher–student finite-model extension	supported	Effective-fidelity lifetime was $2.36\times/1.58\times$ fixed parameters; 95-scalar student retained most teacher gain with bit-exact RTL	Ten-cycle component/model evidence; not LER, wall-clock lifetime, or official GQF superiority
Historical CNN residual LER	conditional	Hybrid Residual-B reduced T24 final LER by 1.75–2.89% relative to UKF in four frozen scenarios	Old mock-backed software-HIL; 0/16 current learned families are same-task eligible
Official GQF superiority	not established	Exact-source checklist 0/15 and all 13 matched metrics null	No reproduction or comparison with Puviani *et al.*
Measured board / external speed	not established	Board latency, jitter and power are null; 18 external rows contain 0 exact same-task comparator	No measured latency, raw-ns ranking, fastest, or SOTA claim

T7.1.4 does not add a new row of system-level superiority to this table. Its positive objects are numerical correctness, validity-domain localization, finite-cutoff channel separation, and fixed-point retention. They strengthen trust in specific components while leaving the V4 LER ordering, V5 early stop, tail non-inferiority, and board-null boundary unchanged.

7 Negative results that define the design

The contract-centric narrative is credible only if its negative evidence is kept visible.

1. **The CNN-centric strong branch failed.** The current main system cannot rely on a learned winner, and the historical CNN does not meet the new matched execution contract.
2. **Static fallback and freeze-all adaptation failed.** All 38 candidate tuples were NO-GO for both V2 and V3 because a frozen trusted state became stale under persistent change. Continuous validated shadows were an evidence-driven correction, not cosmetic refactoring.
3. **Fallback is not free.** The earlier uncertainty gate helped the aggregate OOD mixture but harmed compound OOD and slightly harmed nominal operation. In the later formal matrix, registered non-inferiority passed, but tail fallback/unnecessary signals reached 59–96% and 59–95%.
4. **The smooth primary gate is not a leaderboard.** Route-A passed the locked EWMA contrast but was worse than static and Window MAP, with negative oracle-gap closure and a gain concentrated in periodic drift.

5. **V5 lacked deployable causal headroom.** Most diagnostic oracle gain came from truth-privileged estimation and activation. The expanded action family added only 0.02549%, so implementation was correctly stopped.

6. **Positive auxiliary results do not repair a failed system gate.** Multimode CPD and the two-GKP-CNOT reproduction are useful independent results; using them to relabel V5 as successful would be cross-task leakage.

7. **Hardware evidence remains layered.** CXXRTL, synthesis, P&R, and a six-cycle contract do not imply a bitstream, transport integration, board timing, power, or quantum-device experiment.

8 Terminal verdicts and remaining gates

The main simulation/pre-board promotion decision is already terminal: V4 is restricted and V5 was stopped. Remaining work can improve integrity, reproducibility, or measured-hardware coverage, but it cannot silently change those verdicts.

Table 14: Current terminal states and remaining evidence gates.

Gate	Current requirement	Consequence for writing
V4 final gate (T6.9.3)	Terminal NO-GO; restricted simulator/pre-board draft only	V4 may support a restricted simulator/pre-board systems account, not broad decoder superiority
V5 final gate (T6.15.5)	Terminal `NO_GO_V5_EARLY_HEADROOM_STOP`	No V5 performance or hardware result may be synthesized from V4 artifacts
Auxiliary integrity (T6.19.3)	Terminal `PASS_AUX_COMPARISON_INTEGRITY`: 206 cells, 24/24 gates and mutations, 162 joint regressions	Allows within-lane secondary reporting only; V5 NO-GO and board Blocked remain unchanged
Claim matrix (T7.1.1)	Terminal `PASS_RESTRICTED_PREBOARD_CLAIM_BOUNDARY_MATRIX`: 29 claims, 45 live artifacts, 19/19 gates and mutations	Freezes title/abstract/results/conclusion placement and permits only a restricted pre-board contract paper
Main Figures 1–2 (T7.1.2)	Terminal `PASS_MAIN_FIGURES_1_2_RESTRICTED_PREBOARD_CONTRACT`: 38 bound elements, 16/16 gates and mutations	Freezes the implemented system/safety schematics while keeping V5 Dropped and board measurement null
Main Figures 3–4 (T7.1.3)	Terminal `PASS_MAIN_FIGURES_3_4_RESTRICTED_PREBOARD_RESULTS`: 55 bound rows, 16/16 gates and mutations, 153 parent-chain regressions	Freezes the restricted V4 result and deterministic pre-board boundary while keeping Window/static/oracle-gap negatives, high fallback, V5 Dropped, and 42 board-null fields visible
Supplement S1–S5 (T7.1.4)	Terminal `PASS_SUPPLEMENT_FIGURE_CONTRACT_RESTRICTED_NONRANKING`: 792 bound rows, 17/17 gates and mutations, 13/13 parents, 210 regressions	Adds physics/model, bound, all-seed/OOD, fixed-point, and Phase 6C audit depth without creating a global rank, device-robustness claim, or board result
Real board (T6.9.2)	Bitstream, transport, correctness, latency distribution, deadline miss, resources and power on the named board	Hardware language remains CXXRTL/synthesis/P&R estimate only
Remaining Phase 7 gates	Frozen main figures, prose, supplement, provenance, and final claim audit must consume the T7.1.1 matrix without upgrading its rows	Figure and prose completion may improve presentation, but cannot change the restricted manuscript decision without new registered evidence

9 Discussion

9.1 Evidence-supported contribution

The contribution at this cutoff is a restricted pre-board contract system, not a winning adaptive decoder or a physical QEC experiment. Its scientific value is the separation of admission, causal observability, decoder accuracy, tail intervention, action capacity, state publication, fixed-point execution, and hardware maturity. Each surface can fail independently and remain traceable to an atomic claim. This organization reuses the affine parameter surface, legacy CNN result, teacher/student work, fixed-point pipeline, and FPGA-oriented bank machinery without allowing any one extension to carry an unsupported system-wide claim. The 29-claim matrix and the Methods/Results contracts make that separation machine-checkable.

This contribution is narrower than the original CNN-centric intuition but more defensible. MAP owns the reported logical-error comparisons; HMM events, fallback, and rollback own fail-closed behaviour; the FPGA-facing path owns a deterministic execution contract; CNN and teacher/student modules remain replaceable extensions. The result is therefore an integration and falsification result: the system exposes which layer prevents promotion rather than averaging heterogeneous evidence into a single apparent win.

9.2 When regime awareness helps

The smooth formal experiment illustrates both the value and the limit of regime-aware adaptation. Because EWMA was selected before formal access, the positive paired interval is meaningful for that locked contrast. Because static joint MAP and Window MAP were reported in the same experiment, the data also reject a best-deployable-decoder claim. The periodic family supplies the only Holm-confirmed smooth opportunity, whereas other families show that extra policy complexity can be neutral or harmful. Regime awareness is useful only when the observed history identifies a change early enough and the available action bank contains a better response.

The V5 entry audit makes this condition quantitative. Large gains available to a truth-privileged oracle did not survive the observed-only, strict-causal selector, and the expanded action family supplied far less headroom than the registered continuation threshold. The appropriate response is not to add a larger estimator after seeing the formal result. A future design must first demonstrate prospective identifiability and action-value headroom on a fresh development split; only then should IMM, BOCPD, a posterior-predictive compiler, or a richer learned module be instantiated.

9.3 Safety, performance, and intervention cost

The abrupt/OOD matrix distinguishes bounded safety from performance advantage. V4 passed every registered degradation margin against locked EWMA, including the nominal controls, and therefore did not repeat the earlier compound-OOD catastrophe. Most tail contrasts were nevertheless exact ties, static MAP was substantially better under calibration shift, and fallback or unnecessary fallback signals occupied roughly 59–96% of several tail intervals. These events are not free: they reduce control availability, suppress adaptation, and can keep a conservative bank active long after an anomaly.

Consequently, "fail closed" is an invariant rather than an LER claim. The atomic A/B protocol, CRC/version checks, last-known-good rollback, and recovery hysteresis prevent undefined or partially published actions; they do not imply that every fallback is necessary or that the fallback decoder is optimal. A future tail policy should therefore report avoided and induced errors, false-update and unnecessary-fallback rates, time in each bank, recovery lag, and nominal non-inferiority together with worst-window LER.

9.4 Computational and hardware cost

Table 15 separates costs that were measured in the project from costs that remain outside the evidence layer. Host-side state estimation and image construction are not part of the six-cycle fast path; the FPGA-facing design consumes a versioned, quantized image. CNN/teacher/student training is offline. No PPO, CNN, RNN, HMM, or other model is trained on the target board, and the current HMM inference itself remains a software slow-loop component.

Table 15: Cost inventory at the T7.2.4 cutoff. A measured project cost and a missing physical cost are not interchangeable.

Surface	Available evidence	Cost or uncertainty that remains
Decoder accuracy	Per-round paired LER and directional Pauli counts on frozen simulated traces	No device-calibrated logical channel or measured lifetime
Slow update	A 4000-cycle cadence, finite image/MAC budget, and matched-budget host profiling	Operating-system jitter, transport congestion, and experimental-computer load are not measured
Tail safety	Fallback, false-update, recovery, availability, and fault-path counters	High intervention occupancy can erase practical adaptation benefit
Fast path	Six-cycle, initiation-interval-one contract and one-million-cycle integer/CXXRTL equivalence	No physical clock, CDC, metastability, ADC/DAC, or source-to-action measurement
FPGA resources	Two integrated profiles and three open-tool P&R seeds	No vendor signoff, board utilization, thermal measurement, or power trace
Learning modules	Offline teacher/student and historical CNN component evidence	No same-task learned winner and no board-resident or on-board training
Physical QEC loop	A future procedure and a framed UART/bitstream candidate	The named board, quantum device, transport, timing reference, and all 42 measured fields are absent

Fixed cycles, packed words, mutation-tested faults, and post-route estimates are stronger than a software-only latency sketch, but they remain one evidence layer below a measured control plane. The normalized external FPGA table contains 18 implementations and zero exact same-task comparator. It is therefore valid to report deterministic architecture and target-device feasibility, but not raw-nanosecond superiority, end-to-end QPU latency, zero board deadline misses, lowest power, or fastest/SOTA FPGA decoding.

9.5 External validity and missing physical evidence

The primary simulator is a single-mode square-lattice approximate-GKP syndrome/effective model with protocol-inspired ancilla, readout, leakage, reset, drift, and burst records. Finite-energy, finite-cutoff, noise-transfer, and channel calculations provide task-local cross-checks. These layers do not constitute a calibrated driven open-system model of a particular cavity and transmon, do not reproduce a physical pulse sequence with measured device parameters, and do not consume a prospective real syndrome stream. Synthetic held-out transitions test the registered generators; they cannot establish the frequency, amplitude, duration, co-occurrence, or observability of drift on a real device.

The present per-round LER is therefore a decoder-level simulation endpoint. It is not a measured logical decay rate, a cavity lifetime, a transmon lifetime, or a physical beyond-break-even result. Converting it to a lifetime through a geometric approximation would omit state preparation, syndrome extraction, ancilla faults, leakage, reset, duty cycle, non-exponential decay, and the cost of correction. The project also has no outer-code scaling result and cannot use single-mode data to claim a surface-code threshold, fault-tolerant threshold, or multimode architectural advantage.

Physical-board evidence is independently absent. The bitstream candidate has not been loaded on the named board, no real transport or timestamp boundary has been measured, and all 42 board fields remain null. The project has no microwave, ADC/DAC, cavity, transmon, cryogenic, or quantum-control-chain integration. These are missing experiments, not quantities that can be imputed from CXXRTL or P&R.

9.6 Relationship to prior decoder classes

The evidence does not support a single leaderboard across CI, ML/MAP, direct NN/RL, AQEC, CPD, NMF, and FPGA systems. Relative to static GKP decoding, the current V4 system supplies safe online state publication and a narrow locked-EWMA contrast, but static joint MAP and Window MAP are stronger on the frozen primary benchmark. Relative to general drift-adaptive decoders, it adds an observed-only packet/budget/bank contract and fault-aware tail actions, but does not establish broad average or worst-window superiority. Relative to Puviani NMF, only finite-model teacher/student directionality is available; official GQF source sufficiency and matched lifetime metrics remain blocked. Relative to existing FPGA QEC decoders, the project supplies an integrated pre-board six-cycle/II=1 path, but the task signatures and timing boundaries do not match and the board has not been measured.

The Phase 6C results remain useful precisely because they are kept separate. The multimode posterior-weighted CPD task has a large, consistent LER and tail gain; the two-GKP-CNOT reproduction validates an analog-ML effect; the structured-lattice CPD reproduction is positive over its partial distance range; and AQEC replay is negative. None of these task-local results rescues the stopped V5 single-mode system or supplies a cross-method global rank.

9.7 Failure-informed design priorities

The accumulated negative results suggest a narrower and more productive order for future algorithmic work. First, quantify whether the real observed stream contains causal information about an upcoming regime before expanding the estimator. Second, measure whether a realizable trusted bank or frame action has enough conditional value to change logical outcomes. Third, optimize a cost-aware policy over LER, tail risk, fallback occupancy, recovery lag, host budget, and bank residency instead of optimizing LER alone. Fourth, calibrate and test on chronological, device-separated real records without scenario-wise threshold tuning. Only after these gates should a posterior-predictive or learned replacement be reconsidered.

This order preserves the old dual-loop intuition without making CNN a required winner. A learned module may still improve compression, state estimation, or action proposal, but it must enter through the same observed-only interface, budget, atomic publication, and held-out statistical gates as Window, EWMA, Kalman, or HMM alternatives. Failed families, restarts, cap hits, and null comparisons remain part of the result rather than being removed during model selection.

9.8 Staged path to real data, board measurement, and physical break-even

Table 16 defines a monotonic evidence ladder. A later stage cannot retroactively promote an earlier one, and each stage uses a fresh prospective split or run. Offline device data do not authorize online actuation; board HIL does not establish quantum benefit; and a guarded QPU run does not establish break-even without a matched physical reference.

This final gate is intentionally stricter than reporting simulated LER or a longer fitted lifetime. The corrected and reference experiments must share the device, wall-clock boundary, initialization/readout treatment, and uncertainty model; correction overhead, rejected shots, leakage, and non-exponential decay must remain visible. Until those data exist, the manuscript contains no real beyond-break-even, physical-lifetime, cavity/transmon, on-board-training, or closed-loop quantum-hardware claim.

10 Reproducibility and evidence availability

Every numerical statement in the main figures is regenerated from repository JSON or source-data CSV rather than transcribed from presentation text. The Python entry point is `docs/figures/make_route_a_manuscript_figures.py`. It exports editable SVG, vector PDF, print-resolution LZW-TIFF, review PNG, and a 58-row plotted-data table under `docs/figures/route_a_manuscript_20260720/`. The accompanying manifest records SHA-256, byte size, task-local source path, figure contract, and every output hash. The

Table 16: Prospective transition from the present pre-board system to a physical QEC claim.

Stage	Required action	Promotion evidence
Real-data intake	Bind permission, device revision, quadrature units, cycle semantics, timestamps, missingness, labels/tomography, and immutable chronological splits	Offline observed-only calibration and baseline comparison; no online or lifetime claim
Shadow mode	Run static and adaptive outputs beside the active experiment without changing actions	Prospective coverage, calibration, lag, false-update, fallback, and compute-budget evidence under a frozen policy
Board/HIL	Bind the named board, pinout, transport, source/bitstream hashes, clocks, and timestamp boundaries; stream at least one million cycles	Bit correctness plus core, transport, source-to-action, end-to-end latency, jitter, deadline, resource, thermal, and power distributions; populate the 42 fields
Guarded QPU	Begin with Pauli/phase-frame updates behind a trusted static bank, interlocks, atomic commit, last-known-good rollback, and stopping rules; active displacement requires separate authorization	Prospective safety and logical-channel comparison without silent fallback or post-selection
Physical effectiveness	Compare corrected and matched uncorrected/best-physical references on the same device, cycle definition, duty cycle, and analysis plan	Directional logical channels, leakage and average channel fidelity versus time, with uncertainty and non-exponential alternatives
Break-even	Freeze the physical reference and estimate effective decay rates without reusing confirmatory data for tuning	$G_{\mathrm{QEC}} = \Gamma_{\text{best physical}}/\Gamma_{\text{logical}} > 1$ with a simultaneous confidence lower bound above one

two reused Chinese presentation figures retain their original source-data and generator paths and are labelled as archival interface/model evidence rather than current promotion results. The frozen system/safety figures use the independent Python entry point `docs/figures/make_t7_1_2_main_figures.py`; their 38-element Source Data, artifact hashes, dimensions, editable SVG/PDF, 600-dpi LZW-TIFF, and review PNG are bound by `docs/t7_1_2_main_figure_contract.json`. Visual QA confirmed that the grey/red V5 and board boxes have no incoming production arrow. The frozen result/pre-board figures use `docs/figures/make_t7_1_3_main_figures.py`; their 55-row Source Data binds every plotted value, interval, status, claim identifier, selector, and parent artifact hash to `docs/t7_1_3_main_result_figure_contract.json`. The bundle separates the restricted EWMA contrast from the mandatory Window/static/oracle-gap negatives, and separates the six-cycle clock model and three-seed post-route estimates from the 42 null board fields. The repository-generated T6.19.3 atlas separately binds all 206 auxiliary cells and its SVG/PDF/TIFF/PNG bundle to the terminal integrity artifact. The T7.1.4 supplement then binds 792 records to 13 live parent verifiers through `docs/t7_1_4_supplement_figure_contract.json` and the corresponding Source Data CSV. The Python-only renderer exports four new editable SVG/vector-PDF/PNG/600-dpi-TIFF figures and links, rather than copies, the four T6.19.3 atlas outputs as Supplement S5. Its manifest records 407 editable SVG text nodes, minimum TIFF dimensions of 2952–4389 pixels, and manual visual QA. The 210 parent regressions are intentionally environment aware: 201 use the dependency-complete environment and nine Phase 6C integrity checks use the frozen numerical environment so that a 1.53×10^{-242} McNemar value is not spuriously rejected by an immaterial SciPy last-bit difference.

Randomized comparisons report the independent cluster unit, paired interval, and decision denominator in their captions or tables. Hardware-facing tables retain the evidence layer—fixed-point, CXXRTL, synthesis, P&R estimate, or board measurement—and preserve unmeasured board fields as null. The appendix maps each result to its human-readable report and machine artifact.

11 Conclusion

The current evidence supports the Route-A system as a deterministic, fail-closed way to separate GKP MAP decoding, bounded observed-only adaptation, state publication, optional learning, and pre-board execution. The direct positive evidence is deliberately narrow: a paired smooth contrast against the locked EWMA baseline with a periodic-only Holm discovery, one-million-cycle integer/CXXRTL agreement, and a six-cycle, II=1 architecture that passed three-seed open-tool placement and routing at the 27-MHz contract clock. These results establish traceable integration and deterministic pre-board feasibility, not a universally superior decoder.

The limiting evidence is equally part of the conclusion. Static joint MAP and Window MAP outperform the proposed policy on the frozen primary benchmark; abrupt/OOD results establish locked-EWMA non-inferiority with high intervention cost rather than broad tail improvement; V5 was stopped before implementation; no same-task learned checkpoint is eligible; official GQF/NMF comparison remains blocked; external FPGA implementations provide no exact same-task comparator; and all 42 board-measured fields remain null. Independent CPD, two-GKP-CNOT, AQEC, FPGA-literature, legacy-CNN, and teacher/student results are therefore secondary or historical lanes, not evidence for a global ranking.

Accordingly, this work makes no claim of a calibrated cavity–transmon model, real logical lifetime gain, physical beyond-break-even, board-resident training, measured FPGA speed/power, or closed-loop quantum-device operation. Promotion requires immutable real-data intake, prospective shadow evaluation, named-board HIL, guarded frame feedback, and a matched physical-channel comparison whose break-even confidence lower bound exceeds one. Until then, the endpoint is a restricted simulator/pre-board contract paper; its negative results determine the next experiment.

A Evidence-bounded supplementary figure suite

The five supplementary figures form an audit trail rather than a second leaderboard. S1 localizes numerical and model-validity domains; S2 separates nondeployable recovery bounds, a software cost proxy, and finite-cutoff channel curves; S3 exposes every formal seed and registered OOD lane; S4 couples fixed-point sensitivity to the non-promotion ledger; and S5 preserves the six Phase 6C task signatures without cross-lane ranking. The complete 792-row Source Data file is `docs/t7_1_4_supplement_figure_source_data.csv`.

Table 17: Supplementary figure roles and non-promotion boundaries.

Figure	Evidence object	What it establishes	What it does not establish
S1	Gradient, 65-point host envelope, noise-transfer sweep	Numerical validation and explicit validity/failure domains	Physical convergence, hardware timing, or low-squeezing validity
S2	Petz/SDP, cutoff extension, top-K, six Pauli states	Bound consistency, approximation sensitivity, finite-cutoff channel behaviour	Deployable recovery, synthesized cost, infinite-dimensional convergence, tomography
S3	24 seeds $\times$ 7 methods and complete OOD lanes	Seed dispersion and lane-local degradation are visible	System- or device-level robustness
S4	Fixed-point OAT, four production profiles, nine terminal states	Representation retention and fail-closed non-promotion	Board-measured LER, resources, speed, or power
S5	Six Phase 6C task-signature lanes	Positive, negative, blocked, null, and literature-only states coexist	A global winner or rescue of the stopped V5 branch

Supplement S1 | Numerical validation and explicit model-validity domains

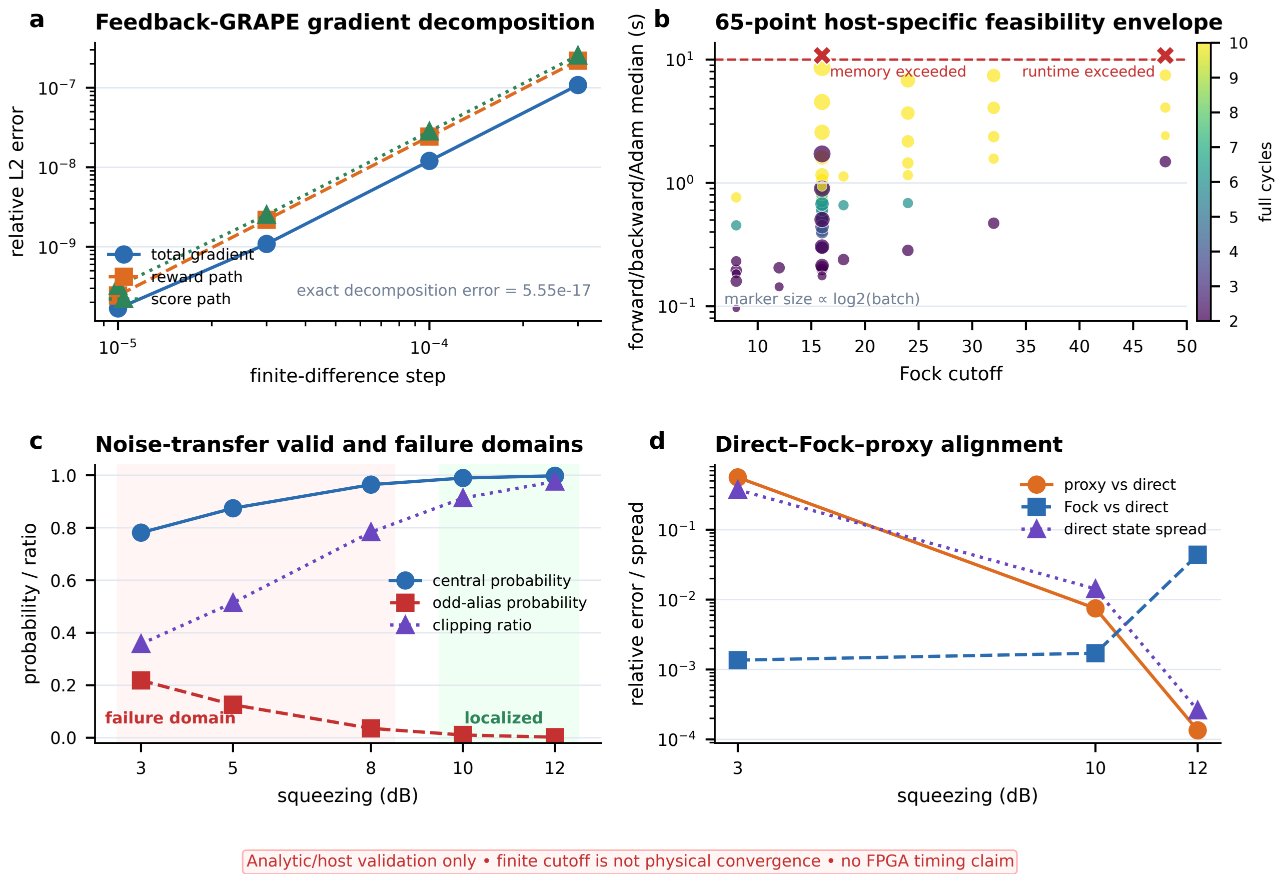


Figure S1: Numerical validation and explicit model-validity domains. (a) The Feedback-GRAPE gradient is checked over the registered finite-difference sweep. (b) Sixty-five cutoff/batch/horizon points expose host memory and runtime boundaries. (c,d) The noise-transfer surrogate is localized at high squeezing and fails at low squeezing. All values are analytic, simulated, or host-resource evidence; no physical-convergence or FPGA-timing claim is implied.

Supplement S2 | Recovery bounds, truncated MAP and six-state channel evidence

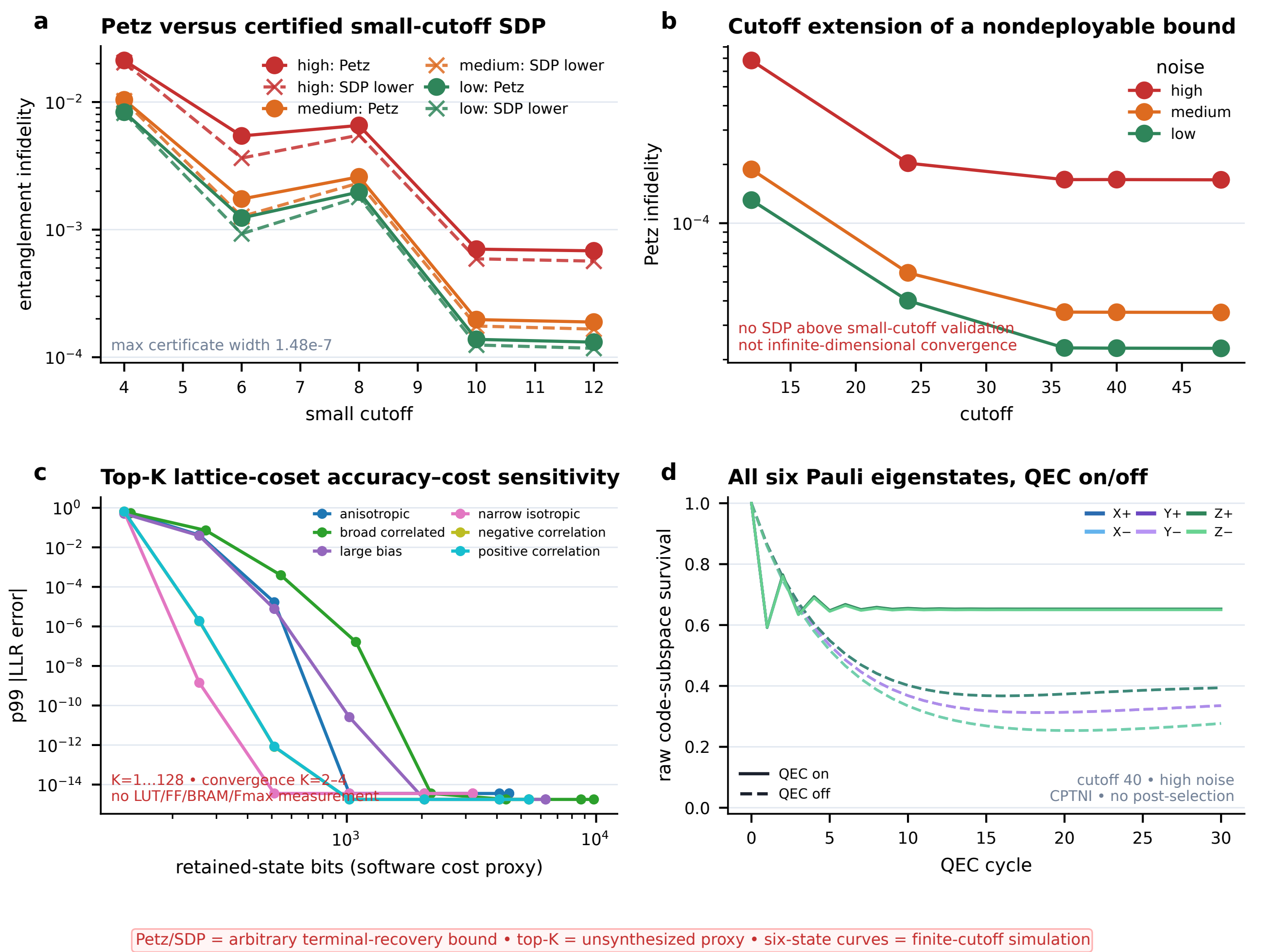


Figure S2: Recovery bounds, truncated MAP, and six-state channel evidence. (a,b) Small-cutoff SDP certificates and the Petz cutoff extension remain arbitrary terminal-recovery bounds. (c) Top-K error is plotted against a retained-state software proxy, not LUT/FF/BRAM or Fmax. (d) QEC-on/off curves cover all six Pauli eigenstates in the finite-cutoff CPTNI simulation without post-selection. The panels are deliberately incomparable to a deployable student/controller or experimental tomography.

Supplement S3 | All formal seeds and complete registered OOD lanes

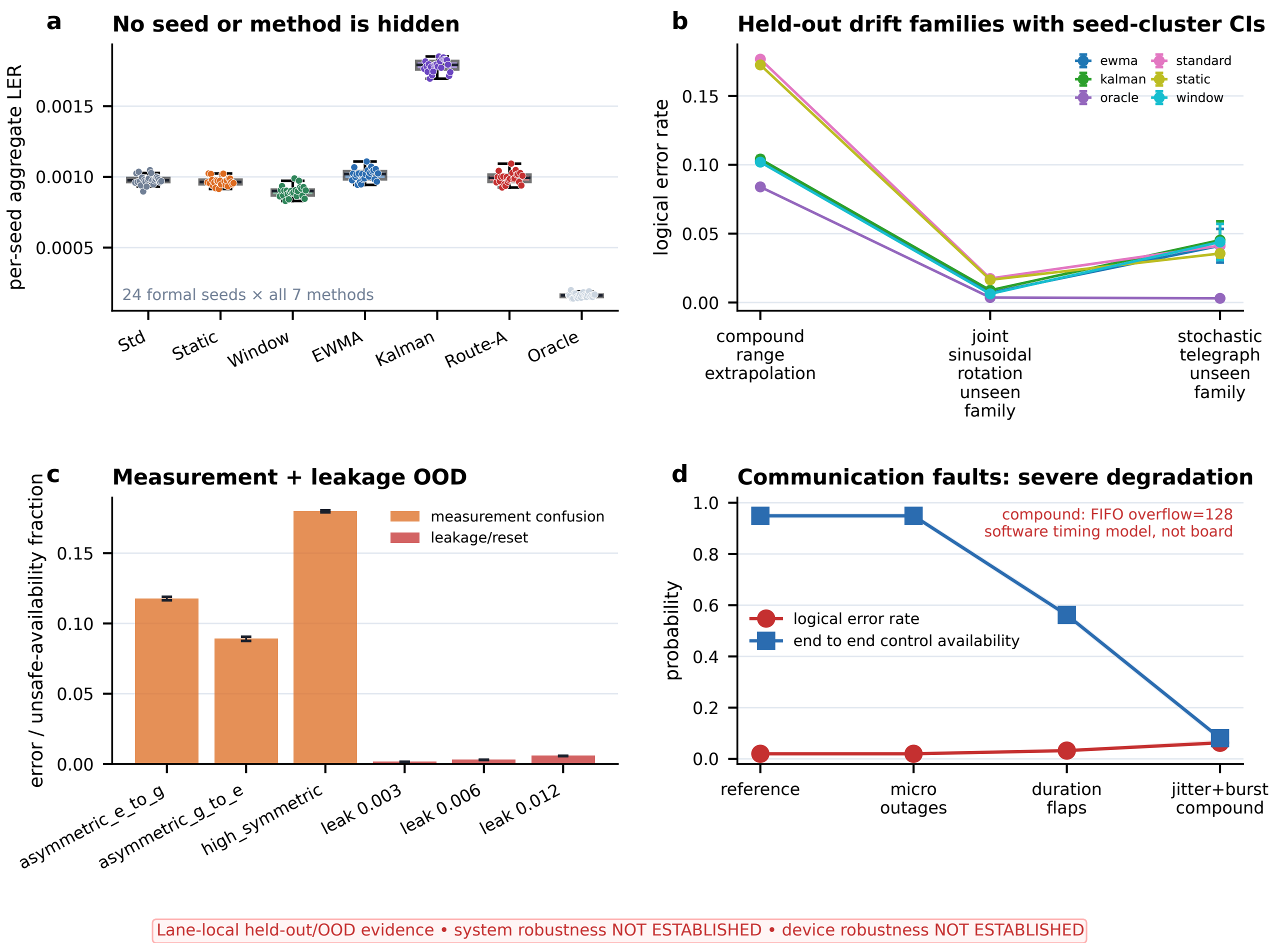


Figure S3: All formal seeds and complete registered OOD lanes. (a) Every one of the 24 seeds for all seven smooth methods is shown. (b–d) Held-out drift, measurement confusion, leakage/reset, and software communication disturbances are reported with their registered cluster intervals. The severe compound communication degradation is a software timing-model result; lane-local OOD does not establish system or device robustness.

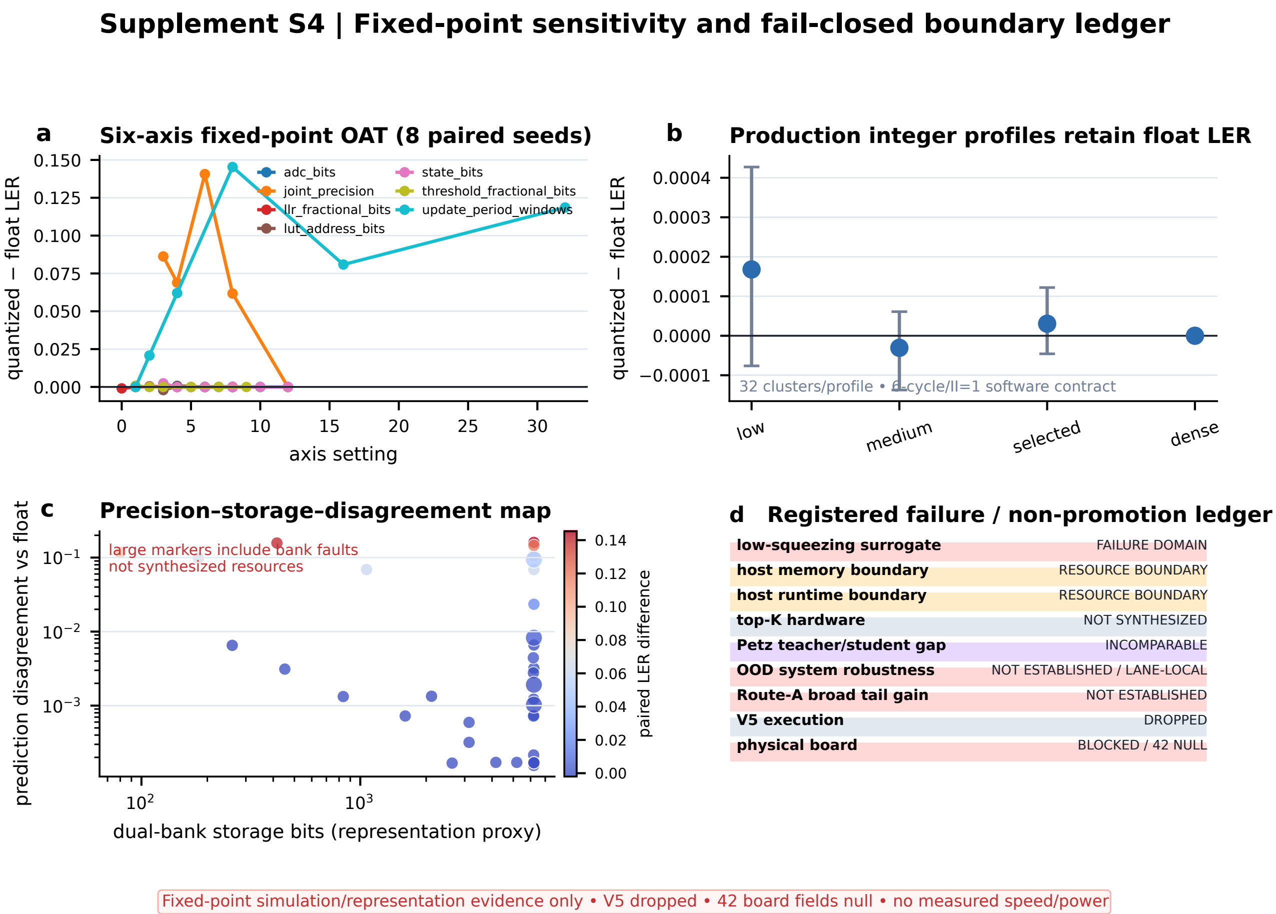


Figure S4: Fixed-point sensitivity and fail-closed boundary ledger. (a) The six-axis OAT retains both benign and sharply degrading settings. (b) Four production integer profiles are compared with float over 32 clusters per profile. (c) Storage and disagreement remain representation proxies rather than synthesized resources. (d) Nine registered failure or non-promotion states keep V5 Dropped and all 42 physical-board fields null.

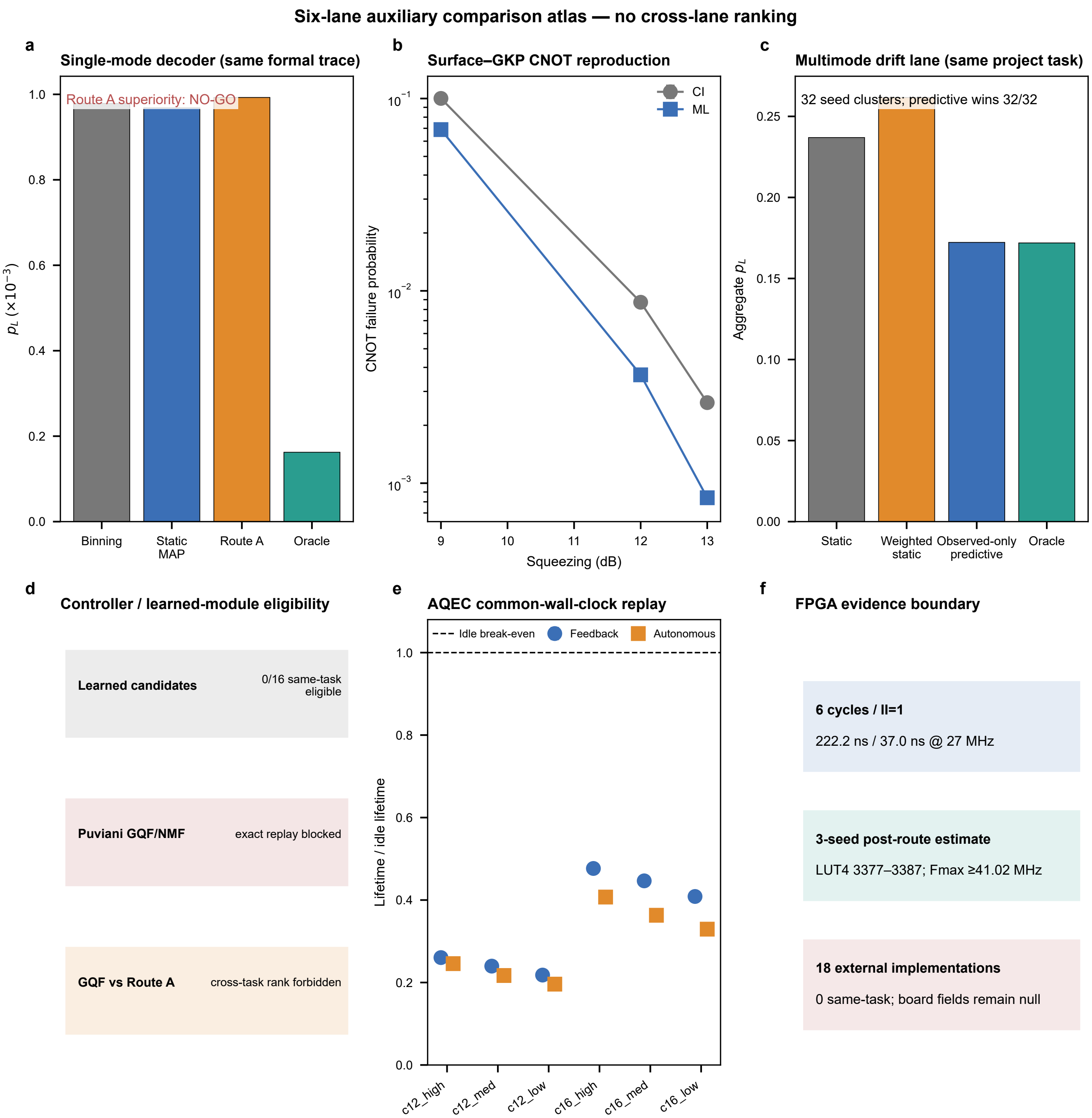


Figure S5: Integrity-gated six-lane auxiliary atlas. The panels retain, rather than hide, a single-mode Route-A NO-GO, a positive CNOT ML reproduction, a positive multimode observed-only CPD result, zero same-task learned/controller eligibility, six negative AQEC wall-clock cells, and FPGA estimate/null boundaries. The 206 cells have no global winner and cannot upgrade the V5 early stop or blocked real-board gate.

B Mathematical and decision definitions

This section fixes the notation used by the executable contracts. It is not a second physical model. Let $L = \sqrt{\pi}$ denote the square-GKP lattice spacing and let $y \in \mathbb{R}$ be one unwrapped scalar syndrome. The folded residue and nearest-cell integer are

$$r_L(y) = y - L\,\mathrm{round}(y/L) \in [-L/2, L/2), \qquad n_L(y) = \mathrm{round}(y/L), \tag{9}$$

with the production quantizer's ties-to-even and saturation rules applied before LUT addressing. Closest-integer decoding returns the parity of $n_L(y)$. Euclidean closest-point decoding is therefore action-equivalent to closest integer only for the registered separable square/isotropic domain; it is not another independent baseline there.

For an observed residue r and parameter state θ, the one-dimensional logical-coset likelihood is

$$\Lambda_b(r;\theta) = \sum_{k\in\mathbb{Z}} p_\theta(r + (2k+b)L)\,, \qquad \widehat{b}_{\mathrm{MAP}} = \arg\max_{b\in\{0,1\}} \Lambda_b(r;\theta). \tag{10}$$

For correlated q/p noise, the strong software joint-MAP comparator instead uses

$$\Lambda_{b_q,b_p}(\boldsymbol{r};\theta) = \sum_{k,\ell\in\mathbb{Z}} p_\theta\Big(\boldsymbol{r} + L(2k+b_q, 2\ell+b_p)^\mathsf{T}\Big)\,, \qquad (\widehat{b}_q, \widehat{b}_p) = \arg\max_{b_q,b_p} \Lambda_{b_q,b_p}. \tag{11}$$

The sums are evaluated with the task-specific frozen truncation or enumerated LUT image. Equation (11) must not be identified with the current phase-conditioned hardware LUT when the correlated likelihood does not factorize.

For N scored rounds, N_a denotes the count whose residual logical Pauli is $a \in \{I, X, Y, Z\}$. The reported channel is

$$p_a = N_a/N, \qquad p_\mathrm{L} = p_X + p_Y + p_Z = 1 - p_I. \tag{12}$$

Non-overlapping windows w contain 512 accepted decisions and have $p_\mathrm{L}^{(w)}$. The p95 is the empirical 0.95 quantile, the worst-window value is $\max_w p_\mathrm{L}^{(w)}$, and CVaR_{95} is the arithmetic mean of the worst five per cent of registered windows, using the task's deterministic tie and endpoint rule. These quantities are finite-horizon simulation summaries, not decay constants or physical lifetimes.

Every paired contrast is oriented as baseline minus candidate:

$$\Delta_B = p_\mathrm{L}(B) - p_\mathrm{L}(\text{candidate}), \qquad G_\text{static} = \frac{p_\mathrm{L}(\text{static}) - p_\mathrm{L}(\text{candidate})}{p_\mathrm{L}(\text{static}) - p_\mathrm{L}(\text{oracle})}. \tag{13}$$

A positive Δ_B favours the candidate. G_static is reported with its sign; a negative value means that the candidate moved away from the oracle relative to static MAP. For a registered fault interval beginning at t_0, detection lag is the first causal declared-fault time minus t_0; recovery lag ends only after the required healthy hysteresis. A fallback is unnecessary only under offline truth-defined nominal membership, and a false update is a committed update inside a preregistered unsafe interval. Truth is used to compute these labels after replay and is absent from every deployable input packet.

C Frozen parameters and execution contract

Table 18 is the parameter registry needed to interpret or repeat the V4 formal lane. A field may be changed only by a new versioned protocol and fresh split; changing it after formal access would create a different experiment.

Table 18: Frozen V4 parameter and execution registry.

Surface	Frozen value	Operational use	Primary owner
Observed packet	Folded q/p syndrome plus typed health, event, integrity, version and age fields	Only online feature surface	T6.5.1–T6.5.2
Truth packet	Separate exact-key simulator state and logical labels	Offline scoring and oracle only	T6.5.2–T6.6.1
Quantization	10-bit ADC, 8-bit LUT address, 2-bit fraction	Common deployable input domain	T6.5.2
MAP word	Signed Q9.12, ties-to-even, saturation	Packed LUT arithmetic	T6.5.2/T6.7.3
Fast path	Five-cycle MAP plus one event/action cycle; II=1	Accepted source word to registered action	T6.5.2/T6.7.3
Posterior cadence	32 decisions	Four-state HMM/event update boundary	T6.6.2
Image cadence	2,048 scalar observations every 4,000 cycles	Completed-window causal proposal	T6.5.2
Equivalent cadence	1,024 paired decisions; 2,000-decision activation period	Comparator alignment	T6.5.2/T6.6.1
Update ceilings	8,192 MAC; 8,192 B state; 8,192 B workspace; 5,000 μs	Admission and deadline accounting	T6.5.2/T6.8.3
HMM states	Normal, smooth, calibration shift, burst	Causal regime posterior	T6.6.2
HMM regularization	0.1	Calibration-only model regularization	T6.6.2
Transition smoothing	4.0	Calibration-only transition prior	T6.6.2
Posterior temperature	2.0	Calibration-only posterior scaling	T6.6.2
Policy tuple	$(0.9, 0.2, 0.25, 192, 2, 8)$	Enter, exit, uncertainty, OOD code, enter and recovery hysteresis	T6.6.3
Smooth promotion floor	0.30 plus lower pre-update predictive NLL than EWMA	Window-bank eligibility	T6.6.2–T6.6.3
Pilot search	1,728 common threshold tuples	One tuple selected before formal access	T6.6.3
Bank transaction	Full inactive image; wire/image CRC, manifest SHA, CAS version, six-cycle drain	Atomic A/B publication	T6.5.2/T6.7.3
Rollback	Monotonic last-known-good republish; eight healthy windows before reopening	Fail-closed integrity recovery	T6.6.2/T6.7.3
Formal split	12 calibration, 12 pilot, 24 formal seed clusters	Model fit, policy selection, untouched inference	T6.5.3
Formal exposure	143 cells; 71,958,528 decisions per method	Smooth plus abrupt/OOD protocol total	T6.5.3/T6.7.1–T6.7.2
Tail window	512 non-overlapping accepted decisions	p95, worst, and action-cost readout	T6.5.3/T6.7.2
Bootstrap	20,000 paired cluster resamples; 95% interval	Seed cluster is the independent unit	T6.5.3

Surface	Frozen value	Operational use	Primary owner
Multiplicity	Holm within each preregistered endpoint family	No cross-lane pooled discovery	T6.5.3/T6.16.3

The parameter image is accepted only after all checks in the same transaction pass. Host acknowledgement without bank, version, epoch, CRC and SHA readback agreement is an incomplete transaction. Deadline and age faults cancel the new candidate; leakage enters the reset FSM; integrity faults permit only LKG republish. No algorithmic branch may convert missing provenance into an ordinary MAP action.

D Complete comparators and statistical protocol

The common-trace baseline set is reproduced in Table 19. "Deployable" here means observed- only and admitted by the common software execution contract; it does not mean that an independent current RTL image exists for every row.

Table 19: Complete comparator registry and ranking role.

Method	State/update rule	Observability	Ranking role	Boundary
Standard binning	Nearest bin on the common quantized grid	Observed-only	Weak reference	Exhaustively checked and compiled to the common LUT domain
Static joint MAP	Fixed full two-dimensional likelihood	Observed-only	Strong software baseline	Not relabelled as current phase-LUT RTL
Window MAP	Completed sliding-window estimate and image	Observed-only	Deployable software comparator	Strongest smooth deployable result in the frozen run
EWMA adaptive MAP	Exponentially weighted estimate and continuously validated shadow	Observed-only	Pilot-locked primary comparator	Only comparator named by the primary V4 hypothesis
Kalman adaptive MAP	Causal state estimate and common image compiler	Observed-only	Secondary deployable baseline	Same trace, precision, cadence and ceilings
V4 ROUTE-A	Four-state HMM/event routing between Window and EWMA shadows plus reset/LKG actions	Observed-only	Registered candidate	Safety policy, not a new MAP likelihood
Hidden-state oracle	Simulator state chooses the corresponding likelihood/action	Truth privileged	Non-ranking upper bound	No deployable, cost, fixed-point or hardware row
Legacy CNN residual	Historical affine residual proposer	Observed history	Matched-budget ablation only	3,489,984 MAC and 1,586,368 B state violate the frozen ceiling

Method	State/update rule	Observability	Ranking role	Boundary
BOCD wrapper	External change detector with matched output interface	Observed-only	External drift diagnostic	LER and 13,004.1-μs worst update must be read together
V5 candidate	Posterior-predictive/risk-aware branch	Intended observed-only	Dropped before implementation	No formal LER, quantized, RTL or P&R result exists

Smooth inference uses four families, six held-out cells per family, 24 formal clusters, 576 trajectories and 28,311,552 scored decisions for every method. The locked abrupt/OOD and nominal matrix adds six fault families, one nominal cell, 24 formal clusters, 888 trajectories and 43,646,976 scored decisions per method. Family aggregates weight the preregistered families equally rather than weighting longer traces more heavily. All seeds, cells, zero-difference families, failed candidate tuples, censored lags and deadline outliers remain in their original denominators.

Paired bootstrap resamples whole seed clusters, never correlated individual rounds. Reported intervals accompany the estimate rather than replace raw counts. Holm correction is applied only within the declared family-specific endpoint set. Non-inferiority margins and catastrophic-degradation thresholds were fixed before formal access. Phase 6C retains the same 95% confidence level and 20,000 paired resamples where applicable, but its p-values and wins are never pooled with Phase 6B. Zero-event hardware/correctness claims use the registered exact one-sided binomial upper bound when a probabilistic statement is made; bit-exact exhaustive or cycle equivalence is otherwise reported as a finite tested-domain statement.

E Ablations, failure modes, and negative selection

Table 20 records executed alternatives and missing branches that constrain interpretation. A negative result is not converted to missing data, and a missing adapter is not converted to a failed algorithm.

Table 20: Ablation and failure-domain ledger.

Object	State	Executed result	Consequence
Static-switch V2	Negative	0/38 pilot-safe tuples passed; closest aggregate slack -0.037109375	A static trusted bank becomes stale under persistent change
Freeze-all-EWMA V3	Negative	0/38 tuples passed; closest aggregate slack -0.044921875	Tail safety cannot mean freezing every estimator
Locked EWMA contrast	Restricted positive	Aggregate paired improvement 2.1687×10^{-5}; only periodic drift survived Holm	Does not establish the best deployable decoder
Static/Window ordering	Mandatory negative	Both beat ROUTE-A on smooth average; static-to-oracle closure is negative	General LER-superiority claim prohibited
Calibration tail	Mandatory negative	Worst window 181/512 for ROUTE-A and 32/512 for static MAP	Safety non-inferiority does not imply tail gain
Fallback occupancy	Mandatory cost	Tail fallback/unnecessary-fallback ranges reach approximately 59–96% and 59–95%	Intervention cost must accompany LER
BOCD matched wrapper	Budget failure	Paired difference 1.00184×10^{-5}, but worst update 13,004.1 μs exceeds 5,000 μs with one miss	Performance cannot be detached from deadline failure

Object	State	Executed result	Consequence
Legacy CNN	Ineligible primary	Historical gains persist, but the current matched MAC/state/workspace budget is violated	Ablation/extension only
V5 causal selector	Negative diagnostic	Nested observed-only headroom -0.2322%; held-out mixture $+0.4587\%$	No estimator implementation without causal headroom
V5 action family	Negative diagnostic	Incremental headroom 0.02549% versus 12% entry gate	No compiler, quantized, RTL or P&R continuation
Teacher/student selection	Qualified extension	Three teacher restarts and nine student fits retained; optimizer cap hits and hindsight reversals disclosed	Finite-model retention, not optimizer optimality
Low-squeezing surrogate	Failure domain	Relative error 0.56037 at 3 dB, versus 7.4845×10^{-3} at 10 dB	High-squeezing localization only
Top-K/Petz	Nondeployable diagnostic	Accuracy/bound and software proxy available; no synthesized comparable hardware	Cannot rank against the deployed student or fast path
Communication compound OOD	Negative software model	LER 0.01986 to 0.06284; availability 0.94899 to 0.08102	Not device robustness or measured transport
Physical board	Blocked/null	42/42 measured fields remain null	No measured latency, jitter, power or deadline claim

F RTL, toolchain, long-sequence, and reproduction procedure

Two one-million-cycle qualifications have different roles and are not merged. T6.2.2 exercised the production fast-path/event FSM over ten 100,000-cycle families: nominal, boundary/frame wrap, leakage/reset, integrity/OOD/stale, deadline pause/recovery, version/commit race, FIFO/backpressure, drop/duplicate/reorder, compound recovery, and saturation. It produced 972,386 valid outputs, zero undefined action, zero output/state CRC error and zero silent overflow. All 61 commit attempts remain visible: 11 acknowledgements and 50 rejections, including ten rollback and ten untrusted rejections.

T6.7.3 then bound the V4 router to frozen formal observed traces. Its ten 100,000-cycle families cover four smooth and six abrupt/OOD regimes. Of the one million cycles, 995,802 are frozen-trace replay and 4,198 are directed boundary/fault vectors. The run contains 31,250 posterior updates, 75 host commit attempts, 25 rollback attempts, 25 commit/config races, 47 tail entries, 28 leakage entries and 59 integrity entries. All visible Python-golden versus CXXRTL fields match, with zero undefined action, CRC error or silent overflow. The directed fraction is 0.4198%; it is disclosed so the result cannot be described as one million independent real traces.

Both qualifications use generated binary trace rows with stored byte counts and SHA-256 digests. T6.2.2 records Yosys 0.67 and MSYS2 g++ 15.1.0, plus CXXRTL model/executable hashes and compile logs. T6.7.3 records its own live source, trace, model, executable and log bindings. The receiver/FIFO model is an abstract bounded disturbance model; CDC, pins, bitstream, analog front end, physical transport and board clock are excluded.

The source-bound static MAP-LUT profile alone is eligible in the Phase 6C hardware table. Its 4,316 accepted equivalence rows have zero action mismatch; the contract is six cycles, II=1, or 222.222 ns and 37.037 ns at an assumed 27 MHz. Three target-device P&R seeds achieve 41.024–42.212 MHz with 3,377–3,387 LUT4, 865 FF, eight BRAM and two DSP in the selected static-core profile. These are open-tool post-route estimates. CI has no independent RTL, V5 was stopped before RTL, and no same-task eligible direct-NN RTL exists; their cells are `N/A`, not zero-cost hardware.

The minimum reproducible sequence is:

1. create the frozen environment named by the task record and verify all input source/config hashes before running an adapter;
2. regenerate the software-golden trace and compare its row count, row width, digest, family coverage and all attempt denominators;
3. regenerate CXXRTL from the bound RTL, compile the recorded harness, and require equality of every visible output plus all injected semantic mutations;
4. run all declared target-device seeds and report the minimum Fmax and every resource count, including failed seeds rather than selecting the best;
5. keep board-only fields null until a named bitstream, physical link, clock, measurement manifest and power/timing capture all exist.

This sequence reproduces project pre-board evidence; it cannot reproduce a physical quantum experiment without the missing device and board inputs.

G Phase 6C source locator and non-mixing ledger

The task signature has 13 fields: code family, modes/distance, decision target, input semantics, history horizon, output action, noise model, observability, online privilege, time basis, compute budget, precision and evidence level. Two numerical values may be ranked only when the decision object and the required signature fields match. Table 21 locates the executable evidence and states the deviation that prevents broader reuse.

Table 21: Phase 6C task signature, source locator, state, and non-mixing rule.

Lane / task	Evidence grade	Executable and machine source	Registered result	Deviation / missing field	Non-mixing rule
Single-mode CI/CPD, T6.17.1	`PROJECT_NATIVE_MATCHED`	`single_mode_cpd_equivalence.py`; T6.17.1 JSON/58-row CSV	Complete q10×q10 domain (1,048,576 points) plus 10^6 boundary samples; zero CI–Euclidean-CPD mismatch	Square/isotropic geometric equivalence only; biased, correlated and finite-energy MAP differ	Do not count CI and equivalent Euclidean CPD twice or rename MAP as CPD
Two-GKP CNOT, T6.17.2	`PROJECT_NATIVE_MATCHED`	`surface_gkp_cnot_reproduction.py`; T6.17.2 JSON/119-row CSV	3,080,192 trials; ML reductions 31.160%, 58.065%, 67.982% at 9/12/13 dB	No official code; 9.9-dB threshold and hardware latency not reproduced	Gate failure is not single-mode memory LER or physical lifetime
Structured CPD, T6.18.2	`OFFICIAL_CODE_REPRODUCTION`	pinned upstream Julia tree; T6.18.2 JSON/1,279-row CSV	2,005/2,005 upstream checks; CPD lower LER in 27/27 independent cells	Partial $d = 3, 5, 7$ grid; no full-distance threshold precision	Threshold and crossing stay in the named surface–GKP construction
Multimode CPD, T6.18.3	`PROJECT_NATIVE_MATCHED`	`multimode_posterior_weighted_cpd.py`; T6.18.3 JSON/CSV	9.6M cycles, 38.4M modal decodes; adaptive $p_L = 0.172261$ vs 0.236929	Independent $d = 3$ balanced heteroscedastic family; not official Lin experiment	Positive LER/tail result cannot promote single-mode V5
Learned or controller, T6.17.3	`INELIGIBLE`	T6.17.3 JSON/CSV	0/16 candidate families pass same-task eligibility	No matching syndrome, action and privilege contract	Literature performance is context, not a result row
GQF/NMF, T6.8.4–T6.8.5	`BLOCKED`	GQF intake/exact-reproduction/gate JSON	0/15 exact-source checks; all 13 matched metrics null	Exact official source/protocol not available in the frozen intake	No claim relative to Puviani NMF

Lane / task	Evidence grade	Executable and machine source	Registered result	Deviation / missing field	Non-mixing rule
AQEC, T6.18.1	`PROJECT_NATIVE_MATCHED`	`aqec_common_wallclock_replay.py`; T6.18.1 JSON/144,152-row CSV	Six cells×24 clusters; active/autonomous minus idle is negative in all cells	Finite-cutoff project replay; official protocol blocked; cycle/us ranking reverses in 144/144 pairs	Control fidelity/lifetime does not rank GKP decoder LER
Project FPGA, T6.19.1	`ESTIMATE_VALUE`	pre-board profiler, RTL, P&R logs; T6.19.1 JSON/3,003-row CSV	Static MAP-LUT six cycles, II=1, three P&R seeds	Board power, jitter, deadline, latency, transfer and commit remain null	Estimate is not a measured speed row
External FPGA, T6.19.2	`LITERATURE_VALUE`	normalized 18-row literature table and source locators	18 implementations; 0 exact same-task comparator	Different code, distance, device, boundary, clock or evidence level	No raw-ns ranking, fastest or SOTA claim
Atlas integrity, T6.19.3	Read-only gate	`secondary_evidence_integrity_gate.py`; 206-cell CSV/JSON	24/24 gates and mutations; no global score or winner	Heterogeneous decision objects are intentionally retained	No averaging, vote counting or rescue of Phase 6B/V5

The value-state vocabulary is lossless. `MEASURED_VALUE` requires a physical measurement at the named boundary; `ESTIMATE_VALUE` denotes simulation, analytic, synthesis or P&R output; `REPRODUCED_VALUE` requires the frozen official-code or project-native protocol; `LITERATURE_VALUE` remains source-scoped. For absent values, `N_A_NOT_APPLICABLE` means that the metric or adapter does not apply, whereas `NULL_NOT_REPORTED` means the field applies but no source-backed value exists. `FAILED` means an attempt ran but failed its frozen gate, and `NEGATIVE` means a valid result disfavouring promotion. The board and GQF records additionally use `blocked` for an unmet external prerequisite and the learned record uses `ineligible` for a candidate whose task signature does not match. None of these states may be replaced by zero or by a neighbouring lane's value.

The final atlas contains 206 cells and no global winner. Its five eligible secondary result objects remain within their own task signatures. The V5 verdict remains `NO_GO_V5_EARLY_HEADROOM_STOP`, and the real-board verdict remains blocked with 42 null measured fields regardless of any positive Phase 6C cell.

H Primary repository evidence map

Topic	Human-readable source	Machine/source-data source
Claim and role contract	`docs/route_a_claim_contract.md`	`docs/t6_5_1_route_a_claim_contract.json` and CSV
Unified execution contract	`docs/unified_execution_contract.md`	`docs/t6_5_2_unified_execution_contract.json` and CSV
Formal preregistration	`docs/route_a_preregistration.md`	`docs/t6_5_3_route_a_preregistration.json` and CSV
Comparator qualification	`docs/unified_comparator_runner.md`	`docs/t6_6_1_unified_comparator_runner.json` and CSV
V4 policy	`docs/regime_aware_safe_policy.md`	`docs/t6_6_2_regime_aware_safe_policy.json` and CSV
Posterior/threshold lock	`docs/route_a_posterior_calibration.md`	`docs/t6_6_3_route_a_posterior_threshold_lock.json` and CSV
Smooth formal matrix	`docs/route_a_smooth_formal.md`	`docs/t6_7_1_smooth_formal_matrix.json` and 498,240-row CSV

Topic	Human-readable source	Machine/source-data source
Abrupt/OOD formal matrix	docs/route_a_tail_formal.md	docs/t6_7_2_abrupt_ood_tail_formal_matrix.json and 686,104-row CSV
V4 integrated qualification	docs/route_a_integrated_rtl_qualification.md	docs/t6_7_3_route_a_integrated_rtl_qualification.json and raw traces
V4 final evidence gate	docs/route_a_final_evidence_gate.md	docs/t6_9_3_route_a_final_evidence_gate.json and source-data CSV
V5 causal headroom / stop	docs/route_a_causal_headroom.md; docs/route_a_v5_final_evidence_gate.md	docs/t6_10_1_causal_headroom.json; docs/t6_15_5_route_a_v5_final_evidence_gate.json
Two-GKP CNOT reproduction	docs/noh_cnot_ci_ml_reproduction.md	docs/t6_17_2_noh_cnot_ci_ml_reproduction_source_data.csv
Multimode CPD	docs/multimode_posterior_weighted_cpd.md	docs/t6_18_3_multimode_posterior_weighted_cpd.json and CSV
Current pre-board profiles	docs/project_preboard_profiles.md	docs/t6_19_1_project_preboard_profiles.json and raw P&R reports
External FPGA refresh	docs/external_fpga_decoder_refresh.md	docs/t6_19_2_external_fpga_normalization.json and CSV
Six-lane auxiliary integrity	docs/secondary_comparison_atlas.md	docs/t6_19_3_secondary_evidence_integrity.json and one-row-per-cell CSV
Paper-level claim matrix	docs/claim_evidence_boundary_matrix.md	docs/t7_1_1_claim_evidence_boundary_matrix.json, 29-row CSV, and verifier
Frozen Main Figures 1–2	docs/main_figures_1_2_contract.md	docs/t7_1_2_main_figure_contract.json, 38-row Source Data, generator, and manifest
Frozen Main Figures 3–4	docs/main_figures_3_4_contract.md	docs/t7_1_3_main_result_figure_contract.json, 55-row Source Data, generator, and manifest
Supplement Figures S1–S5	docs/supplement_figures_contract.md	docs/t7_1_4_supplement_figure_contract.json, 792-row Source Data, bundle manifest, and 13 parent verifiers
Legacy frozen result	docs/paper_materials/paper_simulation_result_table_pack.md	runs/p4_benchmark/T24_formal_software_revalidation_20260510_200743/comparison.csv
Target P&R	docs/target_device_synthesis.md	docs/t5_5_2_target_device_synthesis.json and raw reports
Student Pareto	docs/precision_resource_performance_pareto.md	docs/t5_5_3_precision_resource_pareto.json and equivalence report
Long RTL qualification	docs/long_rtl_qualification.md	docs/t6_2_2_long_rtl_qualification.json and CSV

Topic	Human-readable source	Machine/source-data source
Dual-loop presentation schematic	`docs/figures/ppt_summary_20260716/drift_adaptive_architecture_cn.pdf`	`docs/figures/make_ppt_summary_figures.py`
Teacher–student presentation figure	`docs/figures/ppt_summary_20260716/effective_fidelity_lifetime_cn.pdf`	`docs/t4_4_4_teacher_student_gain_retention_source_data.csv` and figure generator
Evidence-bounded manuscript figures	`docs/figures/route_a_manuscript_20260720/`	Source-data CSV and SHA-256 manifest in the same directory
Auxiliary atlas figure	`docs/figures/t6_19_3_secondary_comparison_atlas.pdf`	T6.19.3 source-data CSV and integrity JSON

I Claim wording guardrails

- Governing contract: each sentence in the title, abstract, results, and conclusion must map to one of the 29 T7.1.1 rows and retain that row's placement, evidence layer, and boundary.
- Allowed now: "improves over the pilot-locked EWMA baseline on the untouched smooth aggregate, with the family-level discovery concentrated in periodic drift."
- Allowed now: "passed the pre-registered abrupt/OOD and nominal non-inferiority gates relative to locked EWMA; most core contrasts were ties."
- Not allowed now: "outperforms static GKP decoding", "best adaptive decoder", "improves tail LER", "outperforms under OOD", or "CNN provides the main gain."
- Required now: state that V5 was stopped before implementation because deployable and incremental action-space headroom failed the entry gates.
- Allowed now: "observed-only posterior-weighted CPD improved aggregate, worst-window, and CVaR metrics in the independent multimode matched task."
- Not allowed now: use the multimode CPD result to claim V5 or full-system promotion, or collapse Phase 6C lanes into a global ranking.
- Allowed now: "the fixed-point core has a six-cycle contract and passed board-independent CXXRTL and target-device P&R qualification."
- Not allowed now: "measured sub-microsecond FPGA decoder", "faster than existing FPGA QEC decoders", or "hardware validated."
- Allowed now: "the distilled student retained most teacher gain in the project finite model and fit a selected pre-board design point."
- Not allowed now: "reproduces or exceeds Puviani NMF" or "proves a universal non-Markovian memory advantage."
- Required now: board latency, jitter and power remain `null`; the six-cycle 222.222 ns value at 27 MHz is a contract-clock estimate.

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